

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

GRADUATION THESIS

Data Collection in Large-Scale WSNs: Designing Algorithms and Simulation

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Program: Global ICT

Supervisor: Associate Professor Nguyễn Khanh Văn

Department: Computer Science

School: School of Information and Communications Technology

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ABSTRACT

Large-scale Wireless Sensor Networks (WSNs) are widely deployed in long-term monitoring applications, where efficient data collection is critical due to the limited energy resources of sensor nodes. As network scale increases, clustering-based data collection schemes suffer from significant scalability issues, particularly the high control overhead induced by frequent cluster head rotation and repeated network topology reconstruction. These challenges are further exacerbated in raw data collection scenarios, where data aggregation is limited and traffic load remains high.

This thesis addresses these limitations by proposing an elastic clustering framework that decouples cluster head rotation from global topology reconstruction through topology reuse and localized reconfiguration. By separating high-level clustering policies from low-level network mechanisms, the proposed framework preserves stable local routing structures across data collection rounds and restricts reconfiguration to affected regions only. This design reduces control overhead, balances energy consumption, and improves scalability without compromising data fidelity.

The proposed framework is quantitatively evaluated using the OMNeT++ simulator with the Castalia framework. Simulation results demonstrate that elastic clustering achieves longer network lifetime, lower control overhead, and more balanced energy consumption compared with representative baseline schemes, including GSTEB and HTC-RDC, especially in large-scale network settings.

In addition, this thesis presents a modular wireless sensor network simulation framework built on top of ns-3. Although not intended for direct performance comparison, the ns-3 based framework provides a flexible and extensible foundation for future system-level research on large-scale and dynamic wireless sensor networks.

Student
Vũ Đức An

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LIST OF ABBREVIATIONS

Abbreviation	Definition
AI	Artificial Intelligence
BS	Base Station
CCO	Cumulative Control Overhead
CH	Cluster Head
DEEC	Distributed Energy-Efficient Clustering
FND	First Node Death, a network lifetime metric
GSTEB	General Self-Organized and Tree-Based routing protocol
HEED	Hybrid Energy-Efficient Distributed clustering protocol
HTC-RDC	Hybrid Tree-based Clustering for Raw Data Collection
IoT	Internet of Things
IP	Internet Protocol
LEACH	Low-Energy Adaptive Clustering Hierarchy
MAC	Medium Access Control layer
ML	Machine Learning
ns-3	Discrete-event network simulator
OMNeT++	Objective Modular Network Testbed in C++
PEGASIS	Power-Efficient Gathering in Sensor Information Systems
PHY	Physical layer
RL	Reinforcement Learning
Sink	Data collection point in a wireless sensor network
UAV	Unmanned Aerial Vehicle
WSN	Wireless Sensor Network

CHAPTER 1. INTRODUCTION

Wireless Sensor Networks (WSNs) have emerged as a fundamental enabling technology for *large-scale monitoring* and data acquisition in a wide range of application domains, including precision agriculture, intelligent transportation systems, and smart environmental monitoring. By deploying a large number of low-cost sensor nodes over extensive geographic areas, WSNs enable continuous observation of physical phenomena and support *data-driven decision making*.

The importance of large-scale WSN deployments has been demonstrated through several real-world systems operating at national and continental scales. In the agricultural domain, the DEMETER project has deployed more than 38,000 heterogeneous sensing devices across 25 pilot sites in 18 European countries to support large-scale agricultural monitoring and decision making [1]. Similarly, in intelligent transportation systems, the Performance Measurement System (PeMS) operated by the California Department of Transportation provides real-time traffic monitoring using nearly 40,000 detectors deployed along major highways and urban areas across California [2].

In contemporary deployments, large-scale WSNs increasingly operate as integral components of broader Internet of Things (IoT) systems, where a massive number of resource-constrained devices must cooperate to support continuous sensing and data-driven services. As a result, many scalability and coordination challenges traditionally studied in WSNs also reflect fundamental system-level issues encountered in modern IoT networks.

In many such applications, WSNs are expected to operate autonomously for long periods of time with minimal human intervention. Consequently, efficient and reliable data collection becomes a critical requirement for system effectiveness, sustainability, and long-term operational success, particularly as network scale continues to increase.

1.1 Problem Statement

As wireless sensor networks scale from hundreds to thousands of sensor nodes, ensuring efficient and reliable data collection becomes increasingly challenging. Large-scale deployments introduce stringent constraints on network lifetime, scalability, and reliability, all of which are strongly influenced by how data is collected and forwarded within the network.

In practical large-scale WSNs, inefficiencies in data collection can quickly lead

to premature node failures, network partitioning, and degraded monitoring performance. These challenges are amplified in applications that require continuous operation and high data fidelity, making scalable data collection a central research problem.

Energy Constraints in Large-Scale WSNs

Sensor nodes in WSNs are typically powered by batteries with limited capacity, and battery replacement or recharging is often impractical in large-scale or remote deployment environments. As a result, *energy efficiency* is a primary design constraint for WSN protocols.

Among all node activities, wireless communication—particularly packet transmission and reception—dominates energy consumption. In large-scale networks, the cumulative energy cost of data forwarding and network coordination increases significantly, making communication efficiency a key determinant of *network lifetime*. Consequently, data collection mechanisms must be carefully designed to minimize unnecessary communication while sustaining long-term operation.

Energy Imbalance and Hotspot Formation

A fundamental challenge in large-scale WSNs is the uneven distribution of traffic load among sensor nodes. Nodes located closer to data sinks or serving as cluster heads are typically responsible for forwarding a disproportionately large volume of packets. This imbalance causes these nodes to deplete their energy much faster than the majority of nodes in the network, leading to the formation of *energy hotspots*.

Once such critical nodes exhaust their energy, network connectivity is severely degraded, potentially isolating large portions of the network and causing premature system failure. Mitigating energy imbalance and preventing hotspot formation are therefore essential for maintaining network reliability and extending operational lifetime.

Control Overhead and Cluster Head Rotation

To address *energy imbalance*, clustering-based data collection has been widely adopted in wireless sensor networks. By organizing nodes into clusters and assigning cluster heads to coordinate data forwarding, clustering reduces long-distance transmissions and enables localized communication.

However, in large-scale networks, cluster heads must be rotated frequently to avoid excessive energy depletion at critical nodes. While cluster head rotation improves energy balance, it often requires repeated reconstruction of network topol-

ogy and routing structures.

In addition to data traffic, these reconfiguration procedures generate a significant amount of management-related communication, including cluster formation messages, cluster head announcements, routing updates, and coordination signaling. The communication cost associated with such network management operations is commonly referred to as *control overhead*. Although control packets do not carry sensed data, they consume bandwidth and energy, and their impact becomes increasingly significant as network scale and operational duration grow.

As a result, frequent cluster head rotation introduces substantial control overhead, consuming additional energy and increasing overall communication load. As network size grows and operational time extends, control overhead becomes a major scalability bottleneck in *clustering-based* data collection schemes.

Raw Data Collection Requirements

The data collection problem is further complicated in applications that require raw or high-fidelity data transmission. In such scenarios, aggressive data aggregation or compression is limited or infeasible, as it may compromise data accuracy or application requirements. Examples include precision agriculture, environmental monitoring, and certain intelligent transportation systems, where detailed sensor readings are essential for reliable analysis and decision making.

Raw data collection significantly increases traffic volume within the network, intensifying energy consumption and exacerbating hotspot and control overhead issues. As a result, data collection mechanisms must efficiently handle high traffic loads while preserving data integrity and maintaining scalability.

Problem Definition

Based on the above considerations, the core problem addressed in this thesis is how to design an efficient data collection mechanism for large-scale wireless sensor networks that supports dynamic cluster head rotation while minimizing *re-configuration overhead*. The objective is to achieve scalable, energy-efficient, and long-lived network operation without sacrificing data fidelity, which remains a critical requirement for many emerging WSN applications.

1.2 Background and Problem Context

This graduation thesis was conducted within the framework of the long-term research project: “*Building Smart Environments on IoT Networks: Addressing Major Algorithmic Challenges*” (acronym: Smart-ENVI). The project is led by Professor Nguyen Khanh Van, who also served as the supervisor for this thesis. The author,

Vu Duc An, is a member of the Smart-ENVI research group, originally assigned to develop algorithms related to the MAC layer.

The thesis, titled “*Data Collection in Large-Scale WSNs: Designing Algorithms and Simulation*”, functions as a constituent project of Smart-ENVI. In this context, the author acted as a key member, responsible for the design and implementation of the MAC layer components, and held primary responsibility for constructing the experimental simulation system for large-scale WSNs.

To ensure completeness and coherence, and with the approval of the supervisor, Prof. Nguyen Khanh Van, this document presents the comprehensive research results of this component project. Within this scope, particular emphasis is placed on the results for which the student author was primarily responsible.

Within the Smart-ENVI project, a primary research direction is the study of data collection mechanisms in large-scale wireless sensor networks, which constitute a core communication substrate for many IoT-oriented systems. The project focuses on clustering-based approaches, which have been widely adopted in WSN research due to their ability to organize large numbers of sensor nodes and improve energy efficiency.

A general observation motivating the project is that much of the existing work on clustering-based data collection has been validated under relatively small or moderate network sizes. When such approaches are considered in larger deployments, new challenges related to network coordination, control signaling, and long-term operation begin to emerge, indicating the need for solutions that explicitly consider scalability at the architectural level.

In parallel, the project recognizes practical difficulties in conducting experimental research on large-scale WSNs. Although several simulation platforms are available, evaluating networks with frequent reconfiguration and long operational timelines remains non-trivial, especially when systematic and repeatable experimentation is required.

Motivated by these observations, the Smart-ENVI project adopts a high-level architectural perspective, exploring data collection solutions that emphasize structural stability and incremental network adaptation. The concept of *elastic clustering* is introduced within this context as a unifying direction, providing the conceptual basis for the research presented in this thesis.

1.3 Research Objectives and Conceptual Framework

This graduation thesis is conducted as a constituent research effort within the Smart-ENVI project, which aims to address fundamental algorithmic and system-level challenges in large-scale wireless sensor networks. At the project level, the overarching objective is to develop scalable data collection solutions that remain energy-efficient and operationally stable as network size and system complexity increase.

Project-Level Research Objectives

From the perspective of the Smart-ENVI project, a central objective is to mitigate the structural inefficiencies observed in clustering-based data collection, particularly those arising from frequent cluster head rotation. While cluster head rotation is essential for balancing energy consumption, existing approaches often incur excessive control overhead due to repeated network-wide reconfiguration.

Accordingly, the project explores architectural solutions that decouple cluster head role changes from full topology reconstruction. The guiding idea is to treat the network as a long-lived system whose structural components can be preserved and incrementally adapted over time. This line of thinking leads to the adoption of the *elastic clustering* concept, which emphasizes topology reuse, localized adaptation, and separation between high-level control decisions and low-level network mechanisms.

Objectives of the Graduation Thesis

Within this broader project context, the specific objectives of this graduation thesis are twofold.

Algorithmic and Protocol-Level Objectives The first objective focuses on algorithmic design at the lower layers of the protocol stack, with particular emphasis on the medium access control (MAC) layer. The goal is to develop coordination mechanisms that can support clustering-based data collection while remaining compatible with network-layer organization.

Rather than introducing a complete cross-layer design, this thesis investigates MAC-level mechanisms that facilitate predictable communication scheduling, efficient local coordination, and reliable interaction with higher-layer routing and clustering decisions. These mechanisms are intended to address practical issues at the interface between local communication behavior and global network structure, which become increasingly significant in large-scale deployments.

Experimental and System-Level Objectives The second objective concerns the construction of a realistic and scalable experimental environment for large-scale wireless sensor networks. A key goal is to enable systematic simulation-based evaluation of networks consisting of hundreds to thousands of nodes, under conditions involving frequent reconfiguration and long-term operation.

To achieve this, the thesis builds upon existing implementations of clustering-based data collection mechanisms developed using the OMNeT++/Castalia simulator, enabling direct comparison with prior work. At the same time, the thesis investigates the migration and reimplementations of these mechanisms within the ns-3 simulation environment, with the aim of overcoming limitations related to extensibility, execution scalability, and long-term research support.¹

Conceptual Framework

Together, these objectives position the graduation thesis at the intersection of algorithmic design and system-level experimentation. The elastic clustering paradigm provides a unifying conceptual framework, guiding both the development of coordination mechanisms and the design of scalable simulation infrastructure.

By aligning local protocol behavior with long-lived network structures, the framework supports adaptive yet stable data collection in large-scale wireless sensor networks. This conceptual foundation directly motivates the architectural and algorithmic design presented in Chapter 3 and informs the experimental methodology adopted in subsequent chapters.

1.4 Contributions

The contributions of this graduation thesis are summarized as follows.

First, with the approval of the project supervisor, this thesis presents the core research outcomes of the Smart-ENVI project related to large-scale wireless sensor network data collection. In particular, it documents the design of *PECEE*, a cellular-based and elastic clustering solution that addresses structural scalability issues caused by frequent cluster head rotation and network-wide reconfiguration. This presentation is intended to provide a coherent and complete account of the project's technical results within an academic thesis context.

Second, the thesis implements and evaluates the proposed elastic clustering mechanisms using the OMNeT++/Castalia simulation framework. A large-scale experimental environment is constructed to support raw data collection and re-

¹A preliminary survey of multiple simulation platforms was conducted during the project, leading to the selection of OMNeT++/Castalia and ns-3 as the primary experimental environments. A detailed discussion of this choice is deferred to Chapter 2.

peated cluster head rotation, and the evaluation demonstrates reduced control overhead, improved energy balance, and extended network lifetime compared with representative baseline approaches.

Third, the thesis designs and implements a modular wireless sensor network simulation framework on top of the ns-3 simulator. The framework introduces WSN-oriented abstractions, configuration-driven scenario specification, and non-IP, data-centric forwarding support, addressing key limitations of existing simulation platforms for large-scale WSN experimentation.

Finally, part of the work presented in this thesis has been peer-reviewed and accepted for publication at an international IEEE conference. The specific technical contributions of the author within this joint research effort are clarified in the accompanying footnote.

1.5 Organization of the Thesis

The remainder of this graduation thesis is organized as follows. Chapter 2 reviews background knowledge and related research on data collection mechanisms in wireless sensor networks, with an emphasis on scalability and control overhead. Chapter 3 presents the problem formulation and the design of the proposed elastic clustering framework. Chapter 4 describes the implementation and quantitative evaluation of the proposed framework using the OMNeT++/Castalia simulation environment. Chapter 5 introduces a modular WSN simulation framework built on ns-3, focusing on architectural design and extensibility. Finally, Chapter 6 concludes the thesis and outlines directions for future research.

¹The work reported in the IEEE WCNC 2026 paper was conducted as a joint research effort. The primary responsibility of the author in this work was the design, implementation, and execution of the simulation-based experimental evaluation, including scenario configuration, performance metric definition, and result analysis. In addition, during the development of the proposed algorithm, the author contributed to several key technical components, notably the cell-level routing mechanism, the intra-cell communication scheduling strategy, and an implicit acknowledgment (implicit ACK) mechanism that significantly reduces ACK control overhead. These contributions were essential to the practical realization and performance evaluation of the proposed framework.

CHAPTER 2. LITERATURE REVIEW

This chapter reviews the research context and related studies relevant to data collection in *large-scale wireless sensor networks*. Building upon the problem formulation and motivation established in Chapter 1, the objective of this chapter is to critically analyze existing work in order to identify the key limitations that motivate the contributions of this thesis.

The chapter first clarifies the scope and assumptions of the research by defining the considered network characteristics, data collection requirements, and operational constraints. These assumptions establish a consistent context for evaluating existing solutions and for positioning the proposed approach.

Subsequently, the chapter reviews representative clustering-based, routing-based, and hybrid data collection mechanisms for wireless sensor networks. The discussion focuses on how these approaches address energy efficiency, scalability, and load balancing, and where they fall short when applied to large-scale deployments and raw data collection scenarios. In particular, the analysis highlights the control overhead introduced by frequent cluster head rotation and topology reconstruction [3].

In addition to algorithmic aspects, this chapter examines commonly used simulation tools for wireless sensor network research. By comparing domain-specific frameworks and general-purpose network simulators, it identifies limitations in current tool support for scalable and extensible WSN-oriented experimentation.

Together, these algorithmic and system-level limitations form the foundation for the elastic clustering framework and the ns-3 based system contribution presented in the subsequent chapters, and establish the conceptual basis for the solution approach developed in Chapter 3.

2.1 Research Scope and Assumptions

This thesis focuses on the problem of data collection in large-scale wireless sensor networks, where a large number of energy-constrained sensor nodes cooperate to support long-term monitoring applications. The research scope is deliberately defined to reflect practical deployment scenarios while maintaining a clear and focused problem boundary.

In terms of network scale, the considered wireless sensor networks consist of hundreds to thousands of sensor nodes deployed over a wide geographic area. Such scale is representative of real-world monitoring systems, including environmen-

tal sensing, precision agriculture, and infrastructure monitoring, where dense node deployment is required to achieve adequate spatial coverage. Sensor nodes are assumed to be static after deployment and to operate autonomously for extended periods of time.

A key assumption of this work is the requirement for *raw data collection*. Sensor readings are forwarded toward data sinks without aggressive in-network aggregation or compression. This assumption reflects application scenarios where data fidelity is critical, such as high-resolution environmental monitoring or data-driven analysis tasks. As a consequence, the network must sustain relatively high traffic volume, especially near cluster heads and forwarding nodes.

Energy efficiency is treated as a primary system constraint. Sensor nodes are powered by limited-capacity batteries, and energy replenishment is assumed to be impractical during network operation. Communication activities, including packet transmission, reception, and *control signaling*, dominate overall energy consumption. Therefore, the research focuses on minimizing communication overhead rather than optimizing local computation or sensing costs.

From a methodological perspective, this thesis restricts its attention to data collection mechanisms based on clustering and multi-hop routing. Issues such as security, fault tolerance, and hardware-level energy optimization are considered outside the scope of this work. In addition, evaluation is conducted through simulation-based experiments, and real-world deployment is not addressed.

In addition, this work assumes a *delay-tolerant data collection model*, where strict real-time delivery is not required. Sensor nodes generate data at a relatively low and stable rate (e.g., one packet per round), allowing the network to prioritize energy efficiency and structural stability over low-latency communication.

By clearly defining these assumptions, this section establishes the context in which existing approaches are evaluated and motivates the need for scalable data collection mechanisms that reduce control overhead and balance energy consumption in large-scale wireless sensor networks.

In summary, the considered scenario is characterized by:

- large-scale deployment with hundreds to thousands of nodes,
- raw and delay-tolerant data collection,
- strict energy constraints and dominant communication cost.

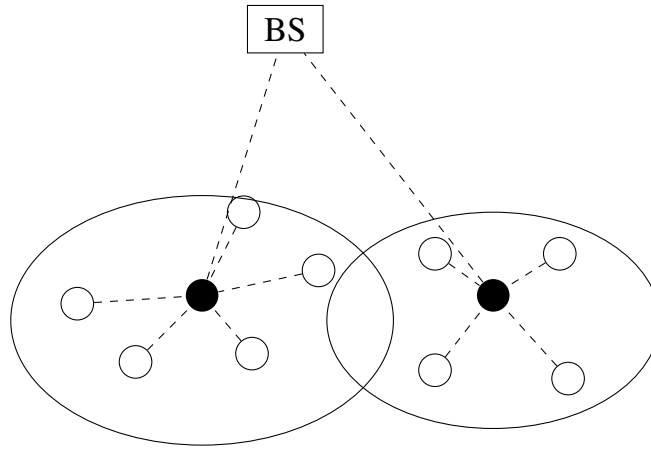


Figure 2.1: Clustering-based data collection structure with intra-cluster and cluster-to-sink communication. *Filled circles denote cluster heads, empty circles denote sensor nodes, and dashed lines represent logical communication paths.*

2.2 Related Research on Data Collection in Wireless Sensor Networks

Data collection has been a central research topic in wireless sensor networks, resulting in a wide range of protocols designed to improve energy efficiency and extend network lifetime. Existing studies can be grouped into several major research directions, including clustering-based data collection, multi-hop routing structures, mobile sink-assisted schemes, and delay-tolerant data collection models. This section reviews these directions with an emphasis on their applicability to large-scale, energy-constrained, and raw data collection scenarios.

Clustering-Based Data Collection: LEACH Family

One of the most influential approaches to data collection in WSNs is clustering-based communication, pioneered by the LEACH protocol. LEACH introduced the concept of organizing sensor nodes into clusters, with periodically rotated cluster heads (CHs) responsible for aggregating and forwarding data to the sink[4]. This design significantly reduces long-distance transmissions and balances energy consumption in small to medium-scale networks (Figure 2.1).

Following LEACH, numerous extensions have been proposed to improve cluster head selection and rotation mechanisms. Distributed schemes such as HEED[5] and DEEC[6] incorporate residual energy and local information to select cluster heads more efficiently, while centralized variants such as LEACH-C leverage global network knowledge to optimize clustering decisions[7]. One representative example is MH-LEACH [8], which enables multi-hop communication among cluster heads to reduce long-distance transmissions to the sink. By allowing cluster heads to forward data through intermediate cluster heads, MH-LEACH improves energy efficiency in larger deployment areas. These protocols demonstrate that CH

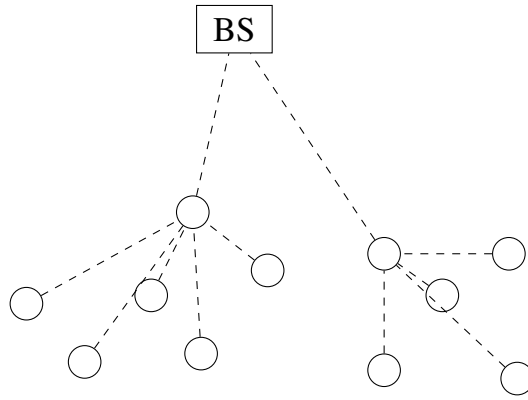


Figure 2.2: Tree-based data collection structure using multi-hop routing toward the sink.

rotation is effective in mitigating energy imbalance at individual nodes.

However, despite their success, LEACH-based approaches typically assume single-hop communication between cluster heads and the sink. As network scale increases, this assumption becomes impractical, leading to rapid energy depletion at cluster heads located far from the sink. Moreover, frequent cluster reformation during CH rotation introduces non-negligible control overhead, which grows with network size.

Multi-Hop Routing and Tree-Based Approaches

To address the limitations of single-hop forwarding, several studies have introduced multi-hop routing structures for data collection. Chain-based protocols such as PEGASIS organize sensor nodes into linear chains, enabling data to be forwarded hop-by-hop toward the sink[9]. While this approach reduces transmission distance, it often suffers from increased delay and poor scalability in large networks.

Tree-based routing protocols, including GSTEB, construct spanning trees rooted at the sink to distribute forwarding load among multiple nodes[10]. These approaches improve scalability and reduce the burden on individual cluster heads. More recent work, such as HTC-RDC[11], further refines this idea by incorporating residual energy awareness and load balancing into the routing structure, specifically targeting raw data collection scenarios in multi-hop WSNs (Figure 2.2).

Despite these improvements, tree- and chain-based approaches typically require global or semi-global topology reconstruction during each data collection round or CH rotation cycle. In large-scale networks, this reconstruction process generates substantial control overhead and energy waste, limiting long-term scalability.

Mobile Sink and UAV-Assisted Data Collection

To further alleviate *energy imbalance* near the sink and improve CH rotation efficiency, mobile sink-based solutions have been proposed. In these schemes, mobile collectors—often implemented using unmanned aerial vehicles (UAVs)—periodically visit selected nodes or cluster heads to gather data directly.

Mobile sink approaches can significantly reduce long-distance forwarding and hotspot formation, thereby extending network lifetime[12]. However, they introduce additional challenges related to mobility planning, scheduling, and coordination overhead. Moreover, most UAV-assisted solutions still rely on underlying clustering or routing mechanisms, meaning that inefficient network reconfiguration remains a critical issue, especially in large-scale deployments.

Common Structural Challenges in Existing Approaches

Across the reviewed approaches, a common limitation becomes evident: energy-efficient data collection and cluster head rotation are typically achieved through repeated reconstruction of network topology. In *large-scale wireless sensor networks*, especially under raw data collection scenarios, the volume of control traffic generated during cluster formation and routing maintenance grows rapidly with network size, significantly reducing network lifetime [13].

More recent studies have explored the use of meta-heuristic optimization techniques to improve clustering efficiency in wireless sensor networks. For example, Maheshwari *et al.* [14] combine butterfly optimization and ant colony optimization to derive energy-efficient clustering and routing decisions. Such approaches demonstrate that global optimization can significantly improve energy distribution under certain conditions.

However, these optimization-based schemes typically rely on complex computation and frequent exchange of global control information to maintain optimality. In large-scale networks with long operational lifetime, the cost of repeated optimization and topology reconstruction can offset the energy gains achieved by optimized clustering, especially in raw data collection scenarios.

These observations raise the question of whether cluster head rotation necessarily requires network-wide topology reconstruction. In existing approaches, the relationship between cluster head rotation and global reconfiguration is often tightly coupled and not explicitly addressed, especially in large-scale networks.

Table 2.1: Structural comparison of representative data collection approaches in large-scale WSNs

Scheme	Approach	Network Size	Ctrl. O/H	Raw Data
LEACH [4]	Single-hop clustering	10–100	High	Limited
LEACH-C [7]		10–100	High	Limited
MH-LEACH [8]	Multi-hop clustering	100–400	Med–High	Limited
HEED [5]		100–400	Med–High	Limited
DEEC [6]		100–400	Med–High	Limited
PEGASIS [9]	Chain / tree-based	200–600	High	Yes
GSTEB [10]		200–800	High	Yes
HTC-RDC [11]	Hybrid clustering tree-based	400–1000	High	Yes
Maheshwari <i>et al.</i> [14]	Optimization clustering-based	100–400	High	Limited
Proposal	Hybrid (elastic clustering)	400–1000+	Low	High

Motivation for Elastic Clustering

The limitations identified in this section motivate the elastic clustering approach proposed in this thesis. By enabling topology reuse and localized adaptation during cluster head rotation, elastic clustering aims to reduce control overhead while maintaining energy balance and scalability. This conceptual framework serves as a bridge between centralized and decentralized designs and forms the foundation of the methodology presented in Chapter 3.

Characteristics summarize

Table 2.1 compares representative data collection schemes with respect to cluster head rotation, topology reconstruction, control overhead, raw data support, and scalability.

Despite differences in clustering strategy and routing structure, a common pattern emerges: most existing approaches rely on network-wide topology reconstruction to support cluster head rotation, resulting in substantial control overhead and limited scalability, particularly under raw data collection scenarios.

2.3 Simulation Tools for Wireless Sensor Networks

Simulation plays a crucial role in wireless sensor network research, as it enables controlled evaluation of data collection mechanisms under diverse network scales, energy constraints, and traffic models. Given the complexity and cost of large-scale

real-world deployments, *simulation-based evaluation* remain the primary means for validating algorithmic designs and system-level concepts in WSNs.

This section reviews representative simulation tools commonly used in WSN research, with a focus on their suitability for large-scale, energy-aware data collection. In particular, it discusses the strengths and limitations of Castalia and ns-3, which form the basis for the experimental evaluation and system contribution presented in later chapters.

Overview of WSN Simulation Tools

Several simulation environments have been developed for wireless sensor network research. OMNeT++ is a general-purpose, discrete-event network simulator that provides a modular architecture and has enabled the development of domain-specific WSN frameworks, most notably Castalia, which extends OMNeT++ with sensor node models, wireless channel abstractions, and detailed energy consumption modeling[15], [16].

Another widely used simulator is ns-3, a general-purpose network simulation platform designed for realism and protocol fidelity. ns-3 provides detailed implementations of networking stacks, wireless communication models, mobility support, and energy frameworks, making it suitable for studying modern and heterogeneous networking systems[17].

While both Castalia and ns-3 can simulate wireless networks, they differ significantly in abstraction level and usability, with important implications for *scalability*, *energy modeling*, and long-term extensibility in large-scale WSN studies.¹

Castalia: Strengths and Limitations

Castalia is a domain-specific simulation framework built on top of OMNeT++ and tailored explicitly for wireless sensor networks. It provides predefined models for sensor nodes, radio communication, MAC protocols, and energy consumption, allowing researchers to focus primarily on algorithmic logic rather than low-level protocol details.

One of Castalia's key strengths lies in its support for *energy-aware WSN research*. The framework includes detailed energy models and explicit differentiation between data traffic and control traffic, making it particularly suitable for studying clustering-based data collection, cluster head rotation, and energy balancing mechanisms. Its modular architecture enables rapid prototyping of routing and clustering algorithms, which has led to its widespread adoption in WSN literature.

¹In this context, abstraction refers to the provision of high-level WSN components—such as sensor nodes, energy models, and routing hooks—that encapsulate low-level simulation details.

However, Castalia also exhibits notable limitations. The framework is no longer actively maintained, and its protocol models rely on simplified abstractions rather than realistic implementations of contemporary networking stacks. Extending Castalia to support emerging research directions—such as large-scale IoT integration, mobile data collectors, or cross-layer optimization—can be challenging and time-consuming. As a result, while Castalia remains effective for algorithm-level evaluation, its suitability for long-term and system-level research is limited.

ns-3 for Wireless Sensor Network Simulation

From a methodological perspective, the choice of simulation tools directly affects not only experimental feasibility but also the scope and longevity of the research outcomes.

An additional methodological consideration in this research concerns the selection of an appropriate simulation platform for large-scale wireless sensor network experimentation. Simulating networks with hundreds to thousands of sensor nodes imposes strict requirements on scalability, energy modeling, and communication abstraction, which makes the choice of a suitable simulator particularly important.

General-purpose network simulators such as OMNeT++ combined with the INET framework provide flexible networking capabilities and realistic protocol implementations. However, when applied directly to large-scale WSN scenarios, these tools offer limited native support for key WSN characteristics, including data-centric communication, non-IP routing, and fine-grained energy modeling. As a result, significant adaptation is often required to evaluate WSN-specific data collection mechanisms.

To address these limitations, this work employs the Castalia framework, which extends OMNeT++ with abstractions specifically designed for wireless sensor networks. Castalia provides detailed energy consumption models and supports clustering-based data collection, making it suitable for algorithm-level evaluation in large-scale WSN scenarios.

At the same time, Castalia exhibits constraints in terms of extensibility and long-term evolution, particularly when considering emerging research directions such as large-scale IoT integration, system-level interaction, and advanced control mechanisms. These considerations motivate the exploration of complementary simulation platforms.

In this context, ns-3 is investigated as an alternative foundation for system-oriented WSN research. Although ns-3 is not originally tailored for wireless sensor

networks, its extensible architecture, active development, and broad model support provide a promising basis for building a reusable and scalable WSN simulation framework. This perspective underlies the ns-3 based framework design presented in Chapter 5.

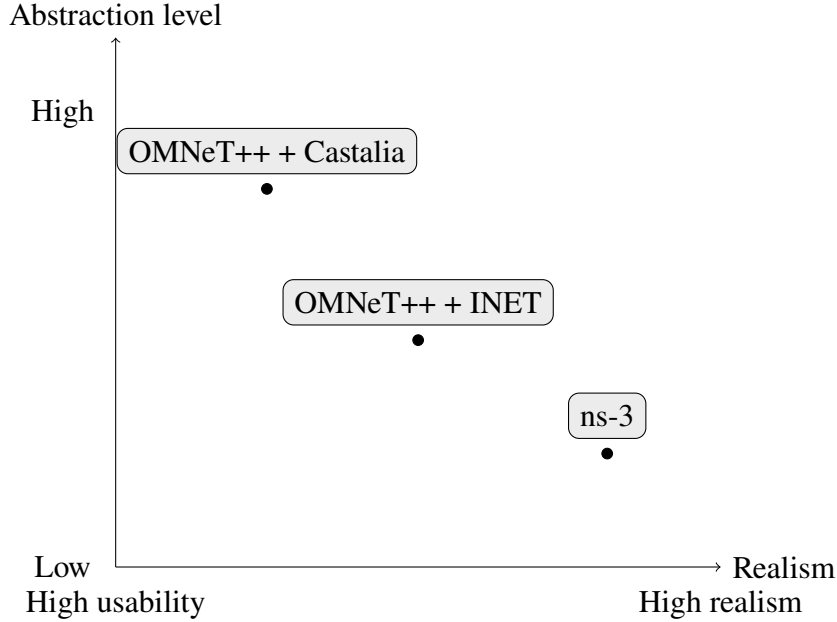


Figure 2.3: While both Castalia and ns-3 are capable of simulating wireless networks, they differ significantly in abstraction level, usability, and suitability for WSN-specific research questions.

In contrast to Castalia, ns-3 is an actively developed, general-purpose network simulator with a strong emphasis on realism and extensibility. It provides detailed models for wireless communication, mobility, energy sources, and protocol stacks, making it a robust platform for studying complex networked systems.

Despite these advantages, ns-3 is not inherently designed for wireless sensor networks. Its architecture is largely *IP-centric*, and it lacks high-level abstractions for common WSN concepts such as data-centric communication, cluster-based organization, and non-IP routing. Consequently, implementing WSN protocols in ns-3 often requires significant manual coding and tight coupling between application logic and networking components.

Nevertheless, the realism and extensibility of ns-3 make it an attractive foundation for future WSN research, particularly as data collection algorithms increasingly interact with IoT systems, mobile sinks, UAV-assisted networking, and AI-based control mechanisms. These trends suggest that ns-3, despite its current limitations, offers greater long-term potential than *domain-specific simulators*.

These considerations suggest that no single simulator fully satisfies the require-

ments of large-scale, energy-aware WSN research, motivating a complementary use of domain-specific and system-oriented tools.

2.4 Performance Metrics and Evaluation Criteria

This section introduces the performance metrics and evaluation criteria commonly used to assess data collection mechanisms in large-scale wireless sensor networks. The purpose of this section is not to present experimental results, but to establish a consistent evaluation perspective that links the research objectives identified in Chapter 1 with the quantitative analysis conducted in Chapter 4.

The selected metrics are chosen to directly reflect the core challenges of large-scale WSNs, including energy efficiency, network lifetime, and the cost of network reconfiguration. These metrics are widely adopted in the literature and provide a meaningful basis for comparing different clustering and routing strategies under energy-constrained and raw data collection scenarios.

2.4.1 Energy Model and Network Lifetime Metrics

Energy consumption is the dominant limiting factor in wireless sensor networks and plays a central role in determining network performance and longevity. In most large-scale WSN studies, communication-related operations—specifically packet transmission and reception—account for the majority of energy expenditure, significantly outweighing sensing and computation costs. As a result, energy models typically focus on accurately capturing communication energy consumption at the node level.

Network lifetime is commonly used as a high-level indicator of energy efficiency and system stability. Among various definitions of network lifetime, the *First Node Death (FND)* metric is widely adopted in the literature. FND is defined as the time or number of data collection rounds until the first sensor node exhausts its energy supply. This metric is particularly relevant in large-scale deployments, as early node failure often indicates severe energy imbalance caused by hotspot formation or excessive forwarding and coordination load.

In addition to FND, residual energy distribution across nodes is frequently analyzed to evaluate how evenly energy consumption is balanced throughout the network. A more uniform residual energy profile generally implies better load distribution and increased robustness against premature network degradation. These energy-related metrics collectively provide insight into the effectiveness of clustering and routing strategies in prolonging stable network operation.

2.4.2 Control Overhead and Reconfiguration Cost

Beyond data transmission, control overhead constitutes a significant and often underestimated source of energy consumption in large-scale wireless sensor networks. Control overhead refers to the communication cost associated with network management operations, such as cluster formation, cluster head rotation, topology reconstruction, and routing maintenance. Unlike data packets, control packets do not carry sensed information but are necessary to maintain correct network operation.

In clustering-based data collection schemes, frequent cluster head rotation can lead to repeated network-wide reconfiguration, generating a large volume of control messages. As network scale increases, this overhead can grow rapidly and become a dominant contributor to energy consumption. Therefore, explicitly accounting for control overhead is essential when evaluating scalability and long-term efficiency.

To capture both instantaneous and long-term reconfiguration costs, cumulative control overhead (CCO) is often employed. CCO is typically defined as the ratio between the total number of control bits transmitted and the total number of transmitted bits over time. This metric provides a normalized measure of how much network capacity and energy are consumed by control operations relative to actual data delivery. A lower CCO indicates a more efficient use of network resources and better suitability for large-scale deployments.

Chapter Summary

This chapter reviewed the research context, related work, and evaluation perspectives relevant to data collection in large-scale wireless sensor networks, building upon the motivation introduced in Chapter 1. The analysis focused on both algorithmic approaches and supporting simulation tools in order to identify the fundamental limitations addressed in this thesis.

The scope of the research was first clarified, emphasizing large-scale WSN deployments with hundreds to thousands of energy-constrained sensor nodes operating under raw data collection requirements. In such scenarios, high traffic load, energy imbalance, and control overhead become critical factors that directly affect network lifetime and scalability.

Existing data collection mechanisms, including clustering-based protocols, multi-hop and tree-based routing approaches, mobile sink and UAV-assisted schemes, and delay-tolerant models, were then reviewed. Although these solutions improve

energy efficiency under certain conditions, a common structural limitation was identified: frequent cluster head rotation typically triggers costly topology reconstruction, resulting in significant control overhead, especially in large-scale and raw data transmission scenarios.

In addition to algorithmic aspects, the chapter examined commonly used WSN simulation tools. The analysis highlights an important limitation at the system level. Domain-specific frameworks such as Castalia provide WSN-oriented abstractions but limited extensibility, whereas ns-3 offers high realism and long-term sustainability but lacks native support for WSN-centric modeling.

Together, these findings highlight two key challenges that motivate the approach proposed in this thesis: the need for a scalable data collection mechanism that decouples cluster head rotation from full topology reconstruction, and the lack of an extensible WSN-oriented simulation framework. These gaps motivate the elastic clustering approach and system-level contributions presented in the subsequent chapters. Based on this foundation, Chapter 3 formulates the problem in detail and introduces the elastic clustering framework as a general solution approach.

CHAPTER 3. PROBLEM FORMULATION AND SOLUTION APPROACH

This chapter formulates the core research problem addressed in this thesis and presents the overall solution approach based on the concept of *elastic clustering*. Building upon the problem context established in Chapter 1 and the analysis of existing research presented in Chapter 2, this chapter transitions from identifying *what* limitations exist in current large-scale wireless sensor networks to explaining *why* these limitations arise and *how* they can be addressed in a systematic and scalable manner.

As discussed earlier, large-scale wireless sensor networks face fundamental challenges in supporting long-term and energy-efficient data collection, particularly under raw data transmission requirements. Although numerous clustering-based and routing-based protocols have been proposed, their performance degrades rapidly as network scale increases. This degradation is not solely the result of inefficient routing paths or suboptimal cluster head selection, but is primarily caused by repeated global reconfiguration of network topology triggered by cluster head rotation and routing maintenance.

From a *system-level perspective*, the root cause of this limitation lies in the tight coupling between cluster head rotation and topology reconstruction in existing designs. In most conventional approaches, changing the role of a relatively small subset of nodes leads to network-wide control message exchange, reformation of clusters, and reconstruction of routing structures. Consequently, control overhead grows disproportionately with network size, consuming a significant fraction of the energy budget and accelerating the formation of energy hotspots.

This chapter argues that improving scalability in large-scale wireless sensor networks requires rethinking the relationship between clustering decisions and network topology management. Rather than treating each data collection round as an independent configuration problem, the network should be viewed as a *long-lived system* whose structural components can be preserved and incrementally adapted over time.

Motivated by this insight, this thesis adopts *elastic clustering* as the central solution approach. Elastic clustering decouples cluster head rotation from full topology reconstruction by reusing previously established local routing structures whenever possible. High-level clustering policies determine which nodes assume coordination roles in each round, while localized network mechanisms adapt existing topologies to accommodate these changes with minimal reconfiguration cost.

By framing elastic clustering as a *conceptual and architectural framework* rather than a single algorithm, this chapter establishes the foundation for the elastic clustering solution developed within the broader team research effort and realized in this thesis. The remainder of the chapter elaborates on the formal problem formulation, design objectives, and architectural principles that guide the algorithmic and system implementations presented in subsequent chapters.

3.1 Problem Formulation and Design Objectives

As established in Chapter 2, data collection in large-scale wireless sensor networks is fundamentally constrained by limited energy resources and highly uneven traffic distribution. In networks consisting of hundreds to thousands of sensor nodes, clustering-based data collection is commonly employed to reduce transmission distance and improve energy efficiency. However, to prevent early energy depletion at cluster heads, these schemes rely on frequent cluster head rotation, which introduces substantial control overhead.

Core Problem Formulation

The core problem addressed in this thesis is the efficient collection of data in large-scale wireless sensor networks under strict energy constraints. In existing clustering-based approaches, cluster head rotation is typically coupled with global or semi-global reconstruction of network topology, including cluster membership and routing paths. As network size increases, the cost of repeatedly rebuilding these structures grows rapidly and consumes a significant portion of the network energy budget.

As a result, control overhead associated with reconfiguration becomes a dominant factor limiting network lifetime, particularly in raw data collection scenarios where data traffic remains high and aggregation opportunities are limited. This overhead not only wastes energy directly but also accelerates energy imbalance by overloading nodes involved in coordination and routing maintenance.

Root Cause Analysis

From a system-level perspective, the root cause of this inefficiency lies in the tight coupling between cluster head rotation and full topology reconstruction. In most existing designs, even a small change in cluster head assignment triggers widespread control message exchange and reinitialization of routing structures across the network. Consequently, the cost of maintaining network organization scales poorly with network size and cluster head rotation frequency.

This design paradigm treats each data collection round as an independent con-

figuration problem, ignoring the fact that large portions of the network structure remain valid across consecutive rounds. The failure to exploit this temporal and structural continuity leads to unnecessary reconfiguration and excessive energy expenditure.

Design Objectives

Based on the above problem formulation and root cause analysis, the design objectives of the proposed approach are defined as follows.

First, the framework should support *dynamic cluster head rotation* while avoiding full topology reconstruction in each data collection round. Second, network reconfiguration should be localized and incremental, such that only nodes directly affected by cluster head changes participate in topology adaptation. This objective aims to minimize control overhead and reduce energy waste associated with coordination traffic. Third, the approach should achieve balanced energy consumption and remain scalable as network size increases, particularly under raw data collection requirements.

Together, these objectives define the need for a data collection framework that decouples clustering decisions from global topology management. This motivation directly leads to the adoption of elastic clustering as a solution approach, which is introduced in the following section.

3.2 Framework Overview and Design Principles

This section presents an overview of the proposed elastic clustering framework and the key design principles that underpin its architecture. Rather than introducing a specific clustering or routing algorithm, the framework provides a structural foundation that enables efficient data collection through *topology reuse* and localized adaptation. The mechanisms introduced here serve as the basis for the concrete rotation and reuse strategies described in subsequent sections.

At a high level, elastic clustering treats the network topology as a reusable and adaptive structure that evolves incrementally across data collection rounds. Instead of rebuilding clusters and routing paths from scratch after each cluster head rotation, the framework preserves stable portions of the network and limits reconfiguration to only those regions directly affected by changes in cluster head assignment. This design directly addresses the control overhead and scalability limitations identified in Chapters 1 and 2.

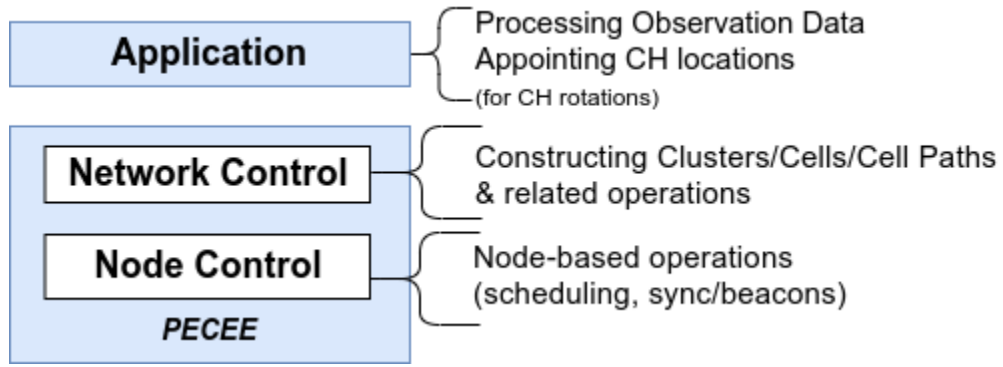


Figure 3.1: Architecture of the proposed elastic clustering framework, illustrating the separation between clustering policy, coordination mechanisms, and node-level communication.

Framework Overview

The elastic clustering framework organizes network operation around a clear separation of responsibilities and a layered architectural structure. Figure 3.1 illustrates the high-level architecture of the framework and the interaction between its main components.

At the top level, the framework distinguishes between clustering policy decisions and network-level mechanisms. High-level decisions, such as cluster head selection and rotation schedules, are treated as external inputs to the framework. These decisions may be based on global information, application requirements, or optimization criteria, but they do not directly dictate how routing and communication structures are constructed.

Below this policy level, the framework provides coordination mechanisms that translate clustering decisions into localized network adaptations. These mechanisms are responsible for maintaining routing structures, managing cluster membership, and enabling topology reuse across rounds. Importantly, the framework is designed such that most network structures persist across cluster head rotations, allowing the network to adapt incrementally rather than through disruptive global reconfiguration.

At the lowest level, node communication and packet forwarding are handled independently of clustering policy. Nodes forward data according to locally maintained routing state, which is updated only when necessary. This layered organization ensures that changes at the policy level do not automatically propagate into costly network-wide updates.

Policy–Mechanism Separation

A central design principle of the elastic clustering framework is the explicit separation between policy and mechanism. In this context, *policy* refers to high-level decisions about network organization, such as which nodes act as cluster heads in a given round. *Mechanism*, in contrast, refers to the concrete procedures by which routing paths, cluster membership, and data forwarding are maintained within the network.

By decoupling these two aspects, the framework avoids embedding specific clustering strategies into the network layer. Cluster head selection can be modified, optimized, or replaced without altering the underlying routing and topology management mechanisms. This separation significantly improves flexibility and makes the framework applicable to a wide range of clustering policies, including centralized, distributed, or hybrid approaches.

From a scalability perspective, *policy–mechanism separation* also enables localized adaptation. When a new cluster head configuration is introduced, the framework does not assume that the entire network must be reconfigured. Instead, only those nodes whose cluster association changes participate in topology updates, while unaffected nodes continue operating using previously established routing structures.

This design principle not only reduces control overhead but also provides a clear architectural pathway for system-level extensions. In particular, it forms the conceptual foundation for the ns-3 based framework introduced in Chapter 5, where clustering policies, routing mechanisms, and communication abstractions are implemented as independent and extensible components.

Summary

In summary, the elastic clustering framework provides an architectural approach for efficient data collection in large-scale wireless sensor networks by emphasizing topology reuse, localized reconfiguration, and separation of concerns. Rather than prescribing a specific algorithm, the framework defines how clustering decisions interact with network mechanisms in a scalable and energy-efficient manner.

The following sections build upon this foundation by detailing the concrete mechanisms for cluster head rotation and topology reuse, and by showing how the framework is applied to the data collection problem in practice.

3.3 Topology Reuse and Cluster Head Rotation Mechanism

This section presents the topology reuse and cluster head rotation mechanism, which constitutes the core contribution of the proposed elastic clustering framework. The mechanism is designed to enable dynamic cluster head rotation while avoiding network-wide topology reconstruction, thereby significantly reducing control overhead and improving energy efficiency in large-scale wireless sensor networks.

Rather than treating each cluster head rotation as an independent re-clustering event, the proposed approach preserves and reuses existing local routing structures. Network adaptation is confined to those regions where cluster head changes actually occur, while the rest of the network continues operating on previously established topologies.

Cellular Overlay Model

To support stable topology reuse and *localized reconfiguration*, the proposed elastic clustering framework organizes the network into a cellular overlay composed of regular hexagonal cells. Each cell acts as a local coordination domain for routing and cluster head rotation.

To prevent mutual radio interference between non-adjacent cells, the edge length of each hexagonal cell is designed to exceed the radio transmission radius. This relationship is controlled by a configurable network parameter $\varepsilon > 1$, which defines the ratio between the cell edge length and the transmission range. By ensuring sufficient spatial separation, simultaneous transmissions in non-adjacent cells do not interfere with each other.

Based on this geometric structure, a minimum of three colors is required to color the hexagonal cell grid such that no two adjacent cells share the same color. In this work, three colors—Red, Green, and Yellow—are used for visualization purposes. As illustrated in Fig. 3.2, adjacent cells are assigned different colors, forming a repeating spatial pattern over the network.

The cell coloring scheme further enables interference-aware scheduling in the time domain. As shown in Fig. 3.3, communication is organized in a TDMA-like manner, where cells of the same color are activated in the same time slot [18], [19]. During a given slot, only nodes belonging to cells of the active color are allowed to transmit, while nodes in other cells remain silent. This scheduling mechanism effectively eliminates inter-cell interference while preserving spatial reuse.

Importantly, the cellular overlay and color-based scheduling remain unchanged

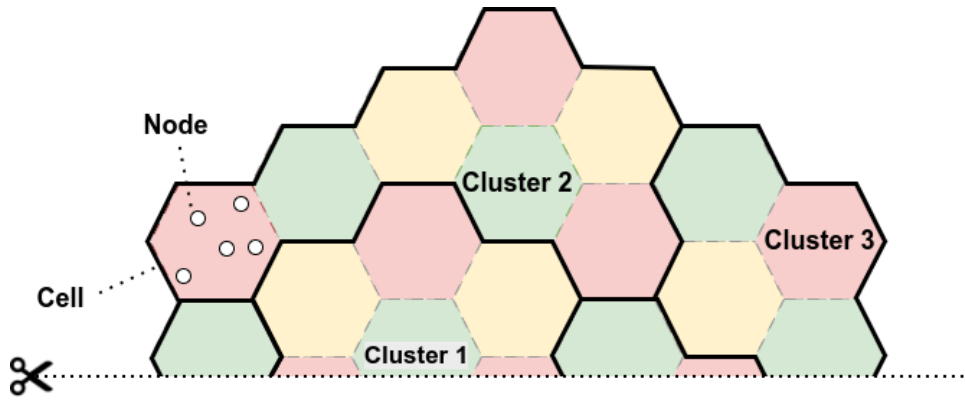


Figure 3.2: The cellular overlay abstraction used to organize the network into localized regions.

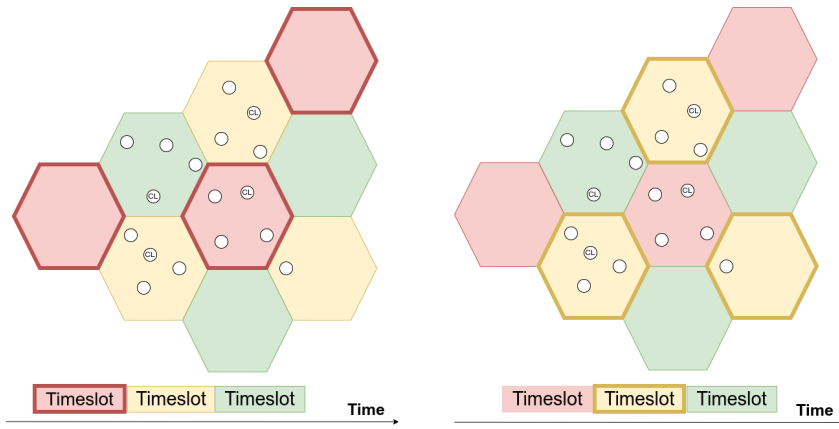


Figure 3.3: Example of constraining transmissions with the Cellular Overlay over the time axis (TDMA)

across data collection rounds. As a result, cluster head rotation and routing adaptation can be performed within a stable structural framework, enabling topology reuse and minimizing control overhead during network reconfiguration.

Within each cell, routing structures are constructed independently and remain largely stable over time. From the perspective of a cell, a change in cluster head assignment does not invalidate internal routing paths. Instead, it only affects how the cell connects to its upstream destination, such as a neighboring cell or a cluster head acting as a gateway.

Figure 3.4 illustrates how cells coordinate routing and forwarding at the inter-cell level while preserving stable intra-cell topology.

This cellular abstraction enables cluster head rotation to be handled as a localized routing adjustment problem rather than a global clustering problem, which is essential for scalability.

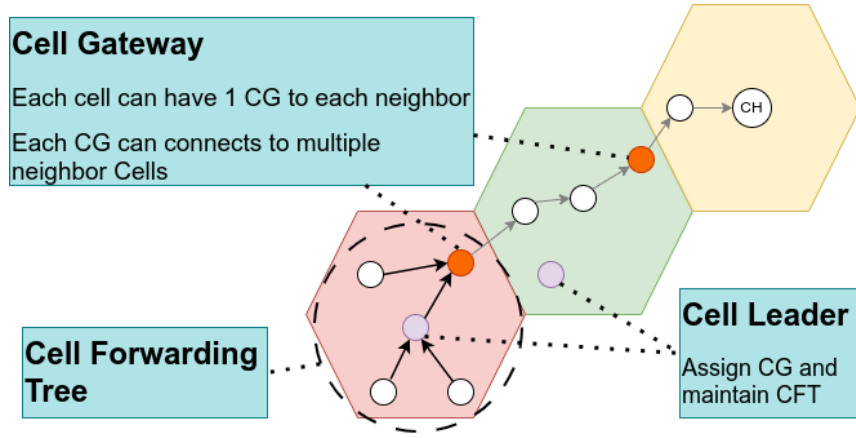


Figure 3.4: Cell coordination and routing based on localized topology reuse.

Intra-cell Topology Initialization

During the initial network setup, each cell independently constructs a set of intra-cell routing topologies toward its neighboring cells. For a given cell, each neighboring cell corresponds to a distinct gateway node located at the cell boundary. A local multi-hop routing tree is built toward each gateway, forming a gateway-centric intra-cell topology.

These routing structures are constructed solely based on local connectivity within the cell and are independent of the global cluster head assignment. Once established, all intra-cell topologies are cached and remain unchanged across data collection rounds. As a result, changes in cluster head assignment do not invalidate internal routing structures, but only affect the selection of the appropriate gateway for inter-cell forwarding.

This initialization phase provides the structural foundation for topology reuse. By decoupling intra-cell routing from cluster head dynamics, the framework enables localized adaptation during cluster head rotation while preserving stable routing within each cell.

Figure 3.5 illustrates the initialization phase of the proposed framework at the cell level. Each cell constructs a set of gateway-centric intra-cell routing topologies, one for each neighboring cell. Once initialized, these topologies are cached and reused across subsequent data collection rounds, forming the basis for efficient topology reuse.

Topology Reuse Mechanism

The fundamental idea behind topology reuse is that, in large-scale wireless sensor networks, most nodes do not need to change their routing behavior when cluster heads are rotated. Instead of discarding the entire routing structure after each data

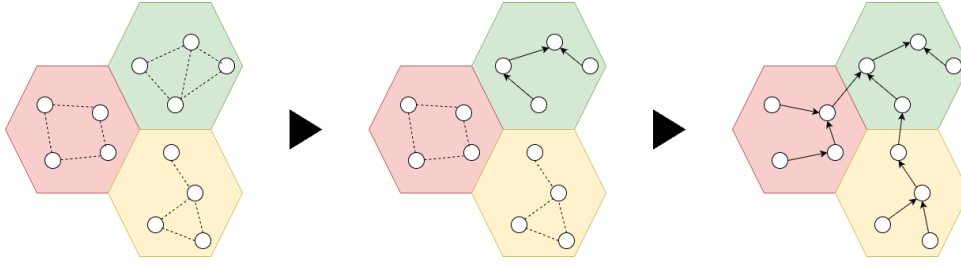


Figure 3.5: Initialization of gateway-centric intra-cell routing topologies. During the initial setup phase, each cell independently constructs and caches multiple routing trees, each rooted at a different gateway toward an adjacent cell. These intra-cell topologies remain stable across data collection rounds and are reused during cluster head rotation.

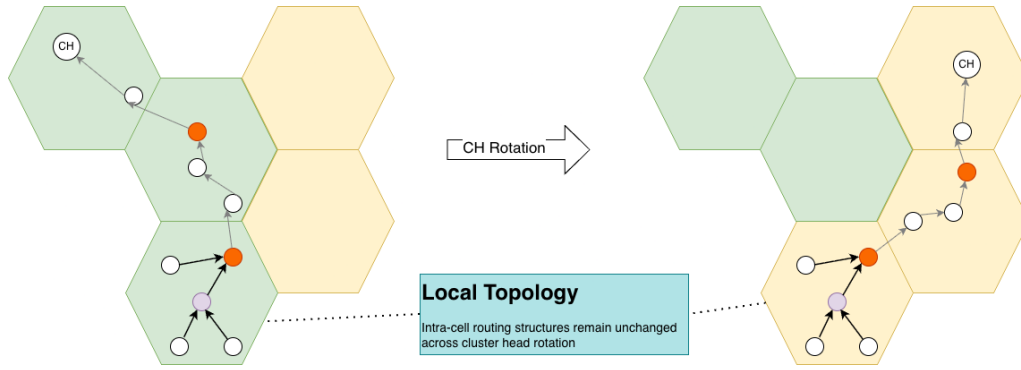


Figure 3.6: Topology reuse mechanism at the cell level. Intra-cell routing structures remain unchanged across cluster head rotation, while only inter-cell forwarding paths toward the active cluster head are updated.

collection round, the proposed framework maintains and reuses local routing state constructed in earlier rounds.

Figure 3.6 and Algorithm 1 together illustrate how topology reuse is achieved through stable intra-cell routing and localized inter-cell adaptation.

When a cluster head rotation occurs, no *intra-cell routing structure* is rebuilt. Instead, each cell determines the neighboring cell that leads toward the new cluster head and activates the corresponding cached topology by selecting the appropriate gateway. Only inter-cell forwarding relationships are updated, while internal routing within each cell remains unchanged.

By confining adaptation to gateway selection and inter-cell coordination, the topology reuse mechanism eliminates network-wide reconfiguration and restricts control overhead to a small, bounded set of cells. This localized adaptation significantly reduces control message exchange, lowers energy consumption, and prevents cascading reconfiguration effects in large-scale wireless sensor networks.

Algorithm 1 Cell-Level Topology Construction and Reuse

Initialization (once)

- 1: **for** $i \leftarrow 1$ k **do**
- 2: Identify gateway GW_{c_i} toward neighboring cell c_i
- 3: Construct intra-cell routing topology T_{c_i} rooted at GW_{c_i}
- 4: Cache topology T_{c_i}
- 5: **end for**
- At each data collection round**
- 6: **if** cluster head assignment unchanged **then**
- 7: Reuse cached T_c without modification
- 8: **else**
- 9: Determine target cell c_{CH} containing the current cluster head
- 10: Select gateway $GW_{c_{CH}}$
- 11: Activate cached topology $T_{c_{CH}}$
- 12: **end if**

Cluster Head Rotation Model

Cluster head rotation is employed to balance energy consumption and prevent premature energy depletion of heavily loaded nodes. In the proposed framework, cluster head selection is treated as a high-level policy decision and is assumed to be performed by a centralized controller, such as the base station, using global information (e.g., residual energy or load distribution).

At the beginning of each data collection round, a new set of cluster heads is announced to the network. Unlike conventional approaches, this announcement does not trigger full re-clustering. Instead, each cell independently evaluates whether its current cluster head or gateway relationship has changed.

From the perspective of an individual cell, cluster head rotation is equivalent to updating the destination of *inter-cell routing paths*. Internal routing within the cell remains unchanged, and previously established paths are reused. This localized view allows each cell to adapt independently, without requiring coordination with unrelated regions of the network.

Cell Path and CHA Packet

To support efficient coordination during cluster head rotation, the framework introduces a lightweight control message referred to as the *Cluster Head Announcement (CHA) packet*. The CHA packet carries essential information about the new cluster head and the associated cell-level routing context.

CHA packets propagate along existing cell paths and are used to update only those routing entries that depend on the cluster head. By reusing established cell

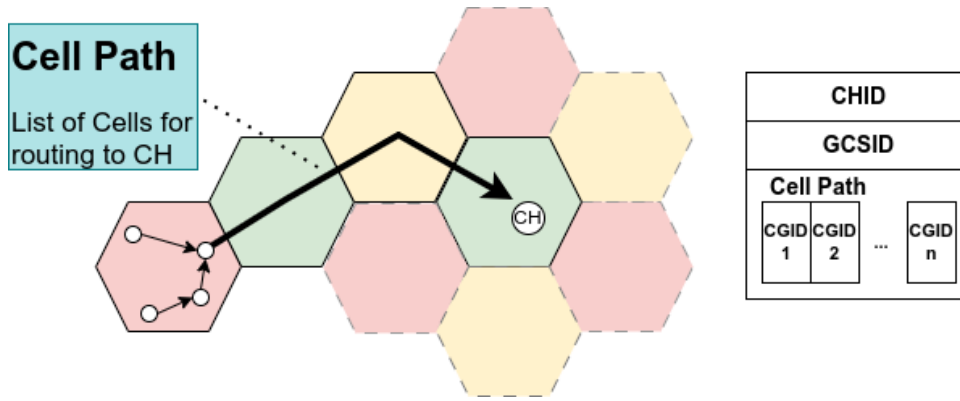


Figure 3.7: Cell path structure and CHA packet used during cluster head rotation.

paths, the dissemination of control information remains localized and efficient.

Cluster Formation Procedure

Algorithm 2 outlines the cluster formation and update procedure executed at the cell level. The algorithm emphasizes reuse of *cached* routing state and minimizes control signaling by restricting updates to affected cells.

Discussion

By decoupling cluster head rotation from full topology reconstruction, the proposed mechanism fundamentally changes how reconfiguration cost scales with network size. Topology reuse ensures that control overhead grows with the number of affected cells rather than the total number of nodes, making the approach particularly suitable for large-scale deployments.

The cellular overlay abstraction further reinforces this scalability by providing a natural boundary for localized adaptation. Together, topology reuse and cell-based rotation form the algorithmic foundation of elastic clustering, enabling efficient, scalable, and energy-aware data collection.

The next section describes how data forwarding is performed over the reused routing structures during the steady-state operation of each data collection round.

3.4 Application to WSN Data Collection

This section describes how the proposed elastic clustering framework is applied to the data collection problem in large-scale wireless sensor networks. The emphasis is placed on the operational workflow of the network and the interaction between configuration, cluster head rotation, and data forwarding, rather than on low-level implementation details.

The proposed framework adopts a round-based operation model, in which network configuration and data transmission are clearly separated. Routing paths used

Algorithm 2 Cluster Formation (executed by each Cell Leader)

Initialization

- 1: $CH_{Aset} \leftarrow \emptyset$ ▷ Set of received CHA packets
- 2: **if** node is selected as cluster head CH_i **then**
- 3: Broadcast $CHA(CH_i)$ to all neighboring cells
- 4: **end if**

CHA Flooding Phase

- 5: **if** $CHA(CH_i)$ is received from neighboring cell leader CL_k **then**
- 6: $CH_{Aset} \leftarrow CH_{Aset} \cup \{CHA(CH_i)\}$
- 7: Rebroadcast $CHA(CH_i)$ to all neighbors except CL_k
- 8: **end if**

Cluster Head Selection Phase

After CHA flooding timeout T_f ▷ Determine the best CH according to the selection policy

- 9: $CH \leftarrow \mathbf{CHSelect-GCS}(CH_{Aset})$
- 10: **if** CH is located in the current cell **then**
- 11: $CG \leftarrow CH$
- 12: $CFT \leftarrow \mathbf{calculateAndCacheCFT}(CG)$
- 13: **else**
- 14: $nextCell \leftarrow$ next cell toward CH (from $CHA.CellPath$) ▷ Reuse cached topology from initial network setup
- 15: $CG \leftarrow \mathbf{getCachedCG}(nextCell)$
- 16: $CFT \leftarrow \mathbf{getCachedCFT}(CG)$
- 17: **end if**

Local Synchronization Phase

- 18: Update CFT information for all cell members

Result

- 19: Each Cell Leader determines the CH, Cell Path, CG, and CFT for the next round

for data forwarding are established during the configuration phase and subsequently reused during the data collection phase, enabling efficient and stable network operation.

Application to WSN Data Collection

In the proposed elastic clustering framework, data packets are forwarded along routing paths determined during the configuration phase. These paths are derived from the trace of the cluster head announcement dissemination, referred to as the *CHAd trace path*. As described in Section 3.3, CHA packets propagate through the cellular overlay to announce cluster head assignments and establish cell-level routing context.

The CHAd trace path implicitly defines a set of forwarding paths from sensor nodes toward their associated cluster heads. By reusing these paths for data trans-

mission, the framework avoids the need for separate route discovery procedures or additional control signaling. As a result, data forwarding becomes a natural continuation of the configuration process.

During the data collection phase, each sensor node generates data packets according to the application sensing rate and forwards them along the pre-established routing paths. In raw data collection scenarios, packets are transmitted without aggregation or modification, preserving data fidelity while leveraging the stability of reused routing structures.

3.4.1 Implicit Acknowledgment in Scheduled Data Forwarding

In conventional wireless communication, explicit acknowledgment (ACK) packets are commonly used to confirm successful packet reception. However, in large-scale wireless sensor networks, frequent ACK exchanges can introduce significant control overhead and additional energy consumption, particularly under multi-hop and raw data collection scenarios.

An alternative approach is *implicit acknowledgment*, where a node infers successful reception by *overhearing* the subsequent forwarding of its transmitted packet by the next-hop node. This mechanism has been explored in prior WSN research as a means to reduce ACK-related control traffic [20], [21].

Implicit acknowledgment is known to exhibit several limitations. In particular, packet loss is not detected immediately, as confirmation is delayed until forwarding activity is observed. Moreover, overhearing requires the receiver to remain active, which can incur additional energy cost if not carefully managed. These characteristics make implicit ACK unsuitable for *latency-sensitive* or unstructured network environments.

In the proposed framework, however, these limitations are consistent with the system assumptions and explicitly addressed by design. The network operates in a round-based and *delay-tolerant* manner, prioritizing energy efficiency and long-term stability over immediate feedback. Furthermore, communication within each cell follows a *predefined schedule*, and routing paths remain stable throughout the data collection phase. As a result, nodes can selectively enable reception during well-defined time slots, enabling reliable overhearing while bounding the energy cost of continuous listening.

By combining scheduled communication, cell-based topology, and stable routing paths, implicit acknowledgment becomes a suitable mechanism for reducing control overhead without compromising correctness. In this context, delayed ac-

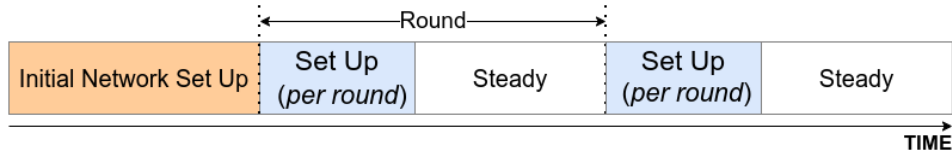


Figure 3.8: Main Phases

knowledge is a deliberate design trade-off that aligns with the operational characteristics of large-scale wireless sensor networks.

3.4.2 Main Phases

The operation of the proposed framework follows a structured and repeatable round-based process, illustrated in Figure 3.8. Each data collection round is composed of three main phases: configuration, cluster head rotation and adaptation, and steady-state data transmission.

- **Configuration Phase:** At the beginning of each round, the base station determines the set of cluster heads and disseminates this information using CHA packets. Cells update their inter-cell routing paths based on the CHAd trace path, while preserving intra-cell topology whenever possible.
- **Cluster Head Rotation and Adaptation:** Cells whose cluster head or gateway changes perform localized routing updates using cached topology information. Cells unaffected by the rotation retain their existing routing state without modification.
- **Data Collection Phase:** Sensor nodes generate and forward data packets along the routing paths established during the configuration phase. Data forwarding is performed in a multi-hop manner, with intermediate nodes acting as simple forwarders. Routing paths remain fixed throughout the data collection phase, ensuring predictable energy consumption and minimal control overhead.

By confining topology adaptation to the configuration phase and maintaining stable routing during data transmission, the framework minimizes control traffic and avoids interference between reconfiguration and data forwarding. This separation of concerns is essential for sustaining long-term operation in large-scale networks.

Overall, the application of elastic clustering to WSN data collection enables efficient reuse of routing structures, reduces coordination cost, and provides a scalable operational model. Together with the topology reuse and cluster head rotation mechanisms presented earlier, this workflow completes the methodological foundation of the proposed framework.

Chapter Summary

This chapter formulated the data collection problem in large-scale wireless sensor networks from a structural and system-level perspective, focusing on the limitations caused by frequent cluster head rotation and repeated network reconfiguration. Rather than treating cluster head selection as the primary challenge, the analysis identified control overhead arising from full topology reconstruction as the root cause of energy inefficiency and poor scalability in existing clustering-based solutions.

Based on this problem formulation, the chapter introduced elastic clustering as a general solution approach. The core idea is to decouple cluster head rotation from global topology reconstruction by reusing previously established local routing structures and limiting network adaptation to affected regions only. This design enables dynamic cluster head reassignment while preserving structural stability across data collection rounds.

The chapter presented the architectural design and guiding principles of the elastic clustering framework, emphasizing policy–mechanism separation, locality of reconfiguration, and scalability through reusable network structures. A topology reuse mechanism and a cluster head rotation model were then described, illustrating how localized adaptations can be achieved using cell-level coordination and CHAd-based routing traces. Finally, the application of the framework to wireless sensor network data collection was detailed through a round-based operational workflow that clearly separates configuration and steady-state data transmission.

Together, these components establish a coherent methodological foundation for efficient and scalable data collection in large-scale wireless sensor networks. Building on this conceptual and algorithmic design, the next chapter presents the implementation of the proposed elastic clustering framework using the OMNeT++ simulator with the Castalia framework, followed by a quantitative evaluation that assesses its effectiveness in terms of network lifetime, control overhead, and scalability.

CHAPTER 4. IMPLEMENTATION AND SIMULATION FRAMEWORK

This chapter presents the implementation and quantitative evaluation of the proposed elastic clustering framework using the OMNeT++ simulator with the Castalia framework. The primary objective of this chapter is to validate the research contributions of the thesis by assessing the effectiveness of topology reuse and localized reconfiguration in large-scale wireless sensor network scenarios.

Building upon the problem formulation and framework design introduced in Chapter 3, the proposed elastic clustering approach is realized within Castalia to enable controlled and repeatable simulation experiments. Castalia is selected as the evaluation platform due to its domain-specific support for wireless sensor networks, including detailed energy consumption models, configurable radio and MAC layers, and native abstractions for clustering-based communication. These features make it well suited for evaluating algorithmic behavior related to energy efficiency, control overhead, and network lifetime.

The evaluation focuses on raw data collection scenarios in large-scale networks, where frequent cluster head rotation and repeated topology reconstruction represent a major source of energy waste in existing approaches. By implementing elastic clustering with topology reuse, this chapter aims to demonstrate how limiting reconfiguration to affected local regions can significantly reduce control overhead, improve energy balance, and extend *network lifetime* as network scale increases.

This chapter is organized as follows. Section 4.1 describes the realization of the proposed elastic clustering logic within the Castalia simulation framework, highlighting how cluster head rotation and topology reuse are implemented on top of existing sensor node models. Section 4.2 presents the simulation configuration and experimental setup, including network parameters, energy models, and traffic patterns. Section 4.3 defines the evaluation metrics and measurement methodology used to assess performance. Section 4.4 discusses the scalability of the proposed framework under increasing network size and cluster head density.

4.1 Implementation and Simulation Setup

This section describes the implementation of the proposed elastic clustering framework and the simulation setup used for quantitative evaluation. The implementation is carried out using the OMNeT++ simulator with the Castalia framework, which provides a domain-specific environment for modeling large-scale wireless sensor networks.

4.1.1 Algorithm Realization in Castalia

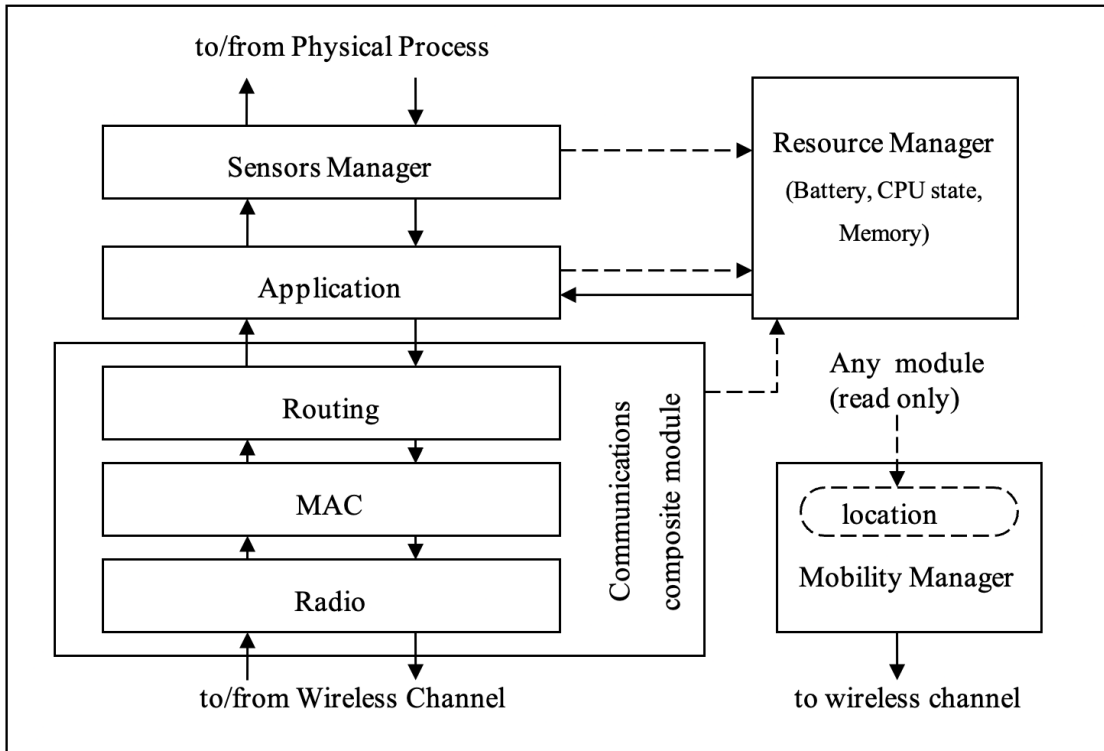


Figure 4.1: Castalia's node composite module

Castalia provides built-in models for sensor nodes, wireless radio communication, MAC protocols, and energy consumption, allowing the implementation to focus primarily on the algorithmic logic of elastic clustering rather than low-level protocol modeling. Each sensor node is modeled using Castalia's standard node architecture, which integrates sensing, communication, and energy modules in a modular and extensible manner.

The realization of the proposed framework concentrates on three core aspects: clustering logic, cluster head rotation, and topology reuse. The clustering logic is implemented as a routing-level component responsible for maintaining cluster membership, forwarding paths, and local routing state. Cluster head rotation is triggered periodically at the beginning of each data collection round based on decisions provided by a centralized controller representing the base station.

Unlike conventional clustering implementations that reconstruct clusters and routing paths from scratch after each rotation, the proposed realization preserves local routing state across rounds. Each node maintains cached neighbor and forwarding information obtained during previous rounds. When a cluster head change occurs, only nodes whose cluster association is affected update their routing paths, while unaffected nodes continue using existing routes. This mechanism directly

implements the topology reuse principle described in Chapter 3 and significantly reduces control message exchange.

Control messages related to cluster head announcements and routing updates are implemented using Castalia message-passing mechanisms. By embedding the elastic clustering logic within Castalia’s routing framework, the implementation ensures that control overhead and energy consumption associated with reconfiguration are explicitly captured by the simulation.

4.1.2 Simulation Configuration

The simulation scenarios are designed to reflect large-scale wireless sensor network deployments and to enable fair comparison with representative baseline schemes. Sensor nodes are randomly deployed over a two-dimensional square area, with network sizes ranging from hundreds to thousands of nodes in order to evaluate scalability. All nodes are assumed to be static after deployment, and a single base station is placed at a fixed location to serve as the final data collection point.

Each sensor node is initialized with the same energy capacity and uses Castalia’s built-in energy model to account for energy consumption during packet transmission, reception, and idle listening. Communication energy dominates overall consumption, while sensing and computation costs are considered negligible. No energy replenishment is allowed during simulation, reflecting battery-powered operation.

Data collection follows a round-based traffic model. In each round, sensor nodes generate data packets at a fixed rate and forward them toward their associated cluster heads. The simulations focus on raw data collection scenarios, where data packets are forwarded without aggregation at intermediate nodes. This traffic model places sustained load on the network and highlights the impact of control overhead and routing efficiency on energy consumption.

Key clustering parameters, including the number of cluster heads, rotation frequency, and control message size, are configured identically across all evaluated schemes to ensure fairness. Each simulation scenario is executed multiple times with different random seeds, and the reported results represent averaged values to mitigate the effects of random node deployment.

Through this implementation and simulation setup, the evaluation isolates the impact of elastic clustering and topology reuse on network performance, providing a controlled environment for quantitative analysis in the following sections.

To ensure clarity and reproducibility, the main simulation parameters used through-

Table 4.1: Simulation Parameters

Parameter	Setting
Initial Energy	0.1 J per node
Node Density	0.0004 nodes/m ²
Sensing Rate	1 <i>pkt/round</i>
Path Loss Model	Free-space

out the evaluation are summarized in Table 4.1. Unless otherwise stated, these parameters are kept identical across all evaluated schemes to guarantee a fair comparison.

4.2 Metric Definition and Measurement Methodology

This section describes the performance metrics used for quantitative evaluation and explains how they are measured within the OMNeT++/Castalia simulation environment. The selected metrics are chosen to directly reflect the core design objectives of the proposed elastic clustering framework, namely energy efficiency, reduced control overhead, and stable long-term network operation.

Network Lifetime

Network lifetime is evaluated using the First Node Death (FND) metric, which is defined as the number of data collection rounds until the first sensor node depletes its energy. FND is widely adopted in wireless sensor network research as an indicator of network stability and energy balance.

In the simulations, each sensor node is initialized with the same energy capacity. Castalia’s energy model tracks energy consumption due to packet transmission, reception, and idle listening. The simulation terminates when the first node reaches zero residual energy, and the corresponding round count is recorded as the network lifetime. This metric highlights the impact of hotspot formation and uneven energy consumption, which are critical issues in large-scale data collection scenarios.

Control Overhead

Control overhead measures the communication cost associated with network management operations, including cluster head announcements, routing updates, and topology adaptation messages. Unlike data packets, control packets do not carry sensed information but are necessary to maintain correct network operation.

In the evaluation, control overhead is quantified by counting the total number of control bits transmitted during network operation. Castalia distinguishes control packets from data packets at the routing layer, allowing precise accounting of control-related communication costs. This metric directly reflects the efficiency of

the cluster head rotation and topology reuse mechanisms.

Cumulative Control Overhead

To capture the long-term impact of reconfiguration costs, cumulative control overhead (CCO) is evaluated. CCO is defined as the ratio between the total number of control bits transmitted and the total number of bits transmitted in the network up to a given round. Formally, it is expressed as:

$$\text{CCO}(r) = \frac{\sum_{i=1}^r B_{\text{control}}(i)}{\sum_{i=1}^r (B_{\text{control}}(i) + B_{\text{data}}(i))},$$

where $B_{\text{control}}(i)$ and $B_{\text{data}}(i)$ denote the number of control bits and data bits transmitted in round i , respectively.

A lower CCO value indicates that a smaller fraction of network resources is consumed by control operations, implying higher efficiency. This metric is particularly important for evaluating the effectiveness of topology reuse, as it reveals how control overhead evolves over successive data collection rounds rather than capturing only instantaneous costs.

Measurement Consistency

All metrics are measured under identical network configurations, traffic models, and energy parameters to ensure fair comparison between the proposed elastic clustering framework and baseline schemes. Each simulation scenario is executed multiple times with different random seeds, and averaged results are reported to reduce the influence of stochastic deployment effects.

Together, these metrics provide a comprehensive and objective basis for evaluating how effectively the proposed framework reduces reconfiguration overhead, balances energy consumption, and extends network lifetime in large-scale wireless sensor networks.

4.3 Simulation Result Analysis

This section presents and analyzes the simulation results obtained from the OMNeT++/Castalia-based evaluation of the proposed elastic clustering framework. The analysis focuses on two primary aspects: network lifetime and control overhead, which directly reflect the design objectives of energy efficiency, scalability, and reduced reconfiguration cost.

Network Lifetime Results

Network lifetime is evaluated using the First Node Death (FND) metric under different network scales and cluster head configurations. Table 4.2 summarizes the

network lifetime results for representative scenarios, including small, medium, and large-scale deployments.

Table 4.2: Network Lifetime for raw data scenario in rounds

Network Settings		Routing Algorithm		
Size	CHs count	PECEE	GSTEB	HTC-RDC
100	3	340	152	253
400	6	174	61	92
400	12	190	88	163
400	24	195	158	173
900	27	178	125	130

To further illustrate the energy balancing effect of elastic clustering, Fig. 4.2 shows the evolution of residual node energy over time for a representative raw data scenario with 400 nodes and 12 clusters.

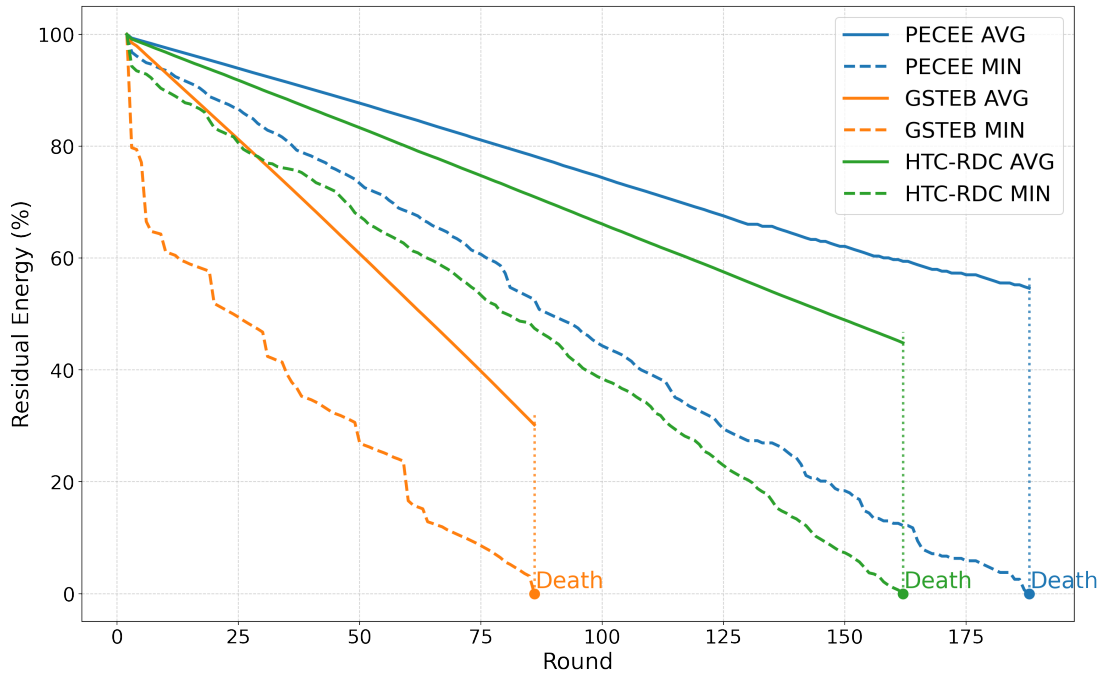


Figure 4.2: Node residual energy over time for raw data scenario in a network of 400 nodes divided into 12 clusters.

The results consistently show that the proposed elastic clustering framework achieves a significantly longer network lifetime compared to baseline schemes such as GSTEB and HTC-RDC. This improvement becomes more pronounced as the network size increases. In small-scale scenarios, the benefit of elastic clustering is already observable due to reduced control overhead. In medium and large-scale networks, where forwarding load and hotspot effects are more severe, the lifetime advantage becomes substantial.

The primary reason for this improvement lies in the reuse of existing routing structures during cluster head rotation. By avoiding full topology reconstruction in each data collection round, the proposed framework reduces unnecessary control traffic and prevents excessive energy depletion at nodes responsible for coordination and forwarding. As a result, energy consumption is more evenly distributed across the network, delaying the occurrence of the first node failure.

These observations confirm that elastic clustering effectively mitigates the energy imbalance problem commonly observed in large-scale wireless sensor networks, especially under raw data collection scenarios where traffic volume remains high and aggregation opportunities are limited.

For completeness, Table 4.3 presents the network lifetime results under data aggregation scenarios.

Table 4.3: Network Lifetime for data aggregation scenario in rounds

Network Settings		Routing Algorithm	
Size	CHs count	PECEE	GSTEB
100	3	480	357
400	12	901	281

Control Overhead Analysis

To quantify the long-term impact of *reconfiguration cost*, the cumulative control overhead (CCO) is analyzed over successive data collection rounds. Fig. 4.3 illustrates the evolution of CCO as a function of round index i for the evaluated schemes.

Control overhead is analyzed by examining both the absolute volume of control traffic and the *cumulative control overhead* (CCO) over successive data collection rounds. The results demonstrate that the proposed framework significantly reduces control overhead compared to baseline approaches.

In conventional clustering-based schemes, cluster head rotation typically triggers network-wide reconfiguration, resulting in a large number of control packets exchanged in each round. In contrast, the elastic clustering framework limits reconfiguration to only those nodes affected by cluster head changes. Nodes that retain their cluster association continue to use existing routing paths without modification, thereby avoiding redundant control messaging.

As a consequence, the cumulative control overhead grows at a much slower rate in the proposed framework than in baseline schemes. This reduction in control traffic directly translates into lower communication energy consumption and

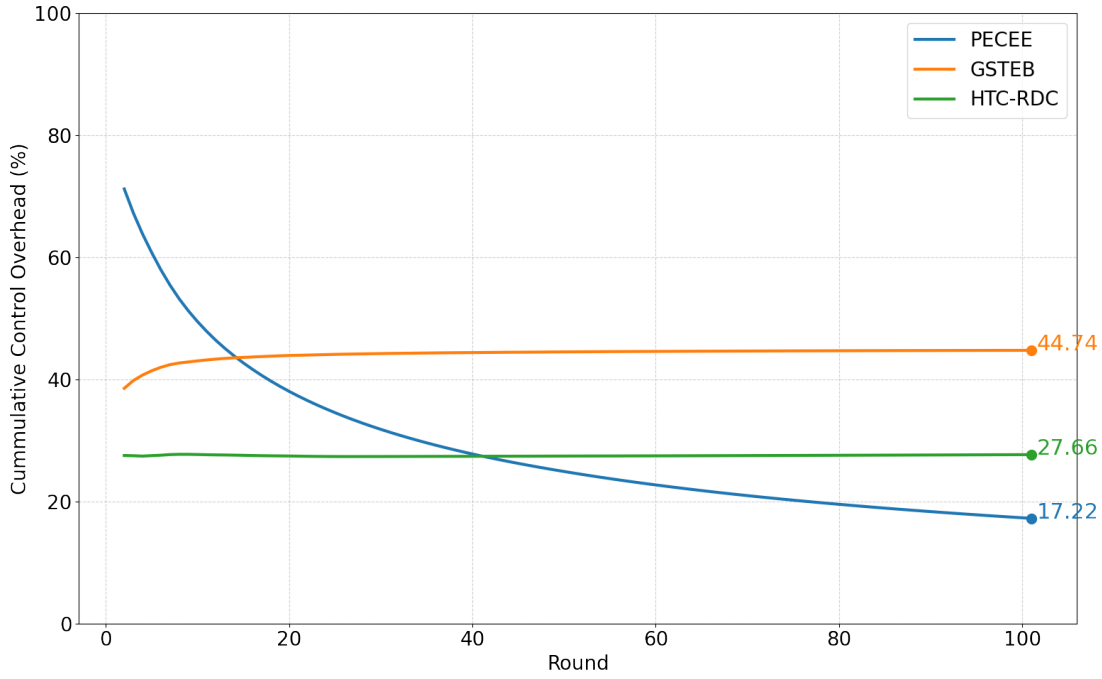


Figure 4.3: Cumulative Control Overhead at round i .

improved long-term network efficiency. Moreover, reduced control overhead alleviates channel contention and interference, indirectly contributing to more stable data delivery.

Overall, the control overhead analysis highlights topology reuse as a key contributing factor to the superior performance of the proposed framework. By minimizing unnecessary reconfiguration, elastic clustering achieves both immediate energy savings and long-term scalability benefits, which are essential for large-scale wireless sensor network deployments.

4.4 Scalability Discussion

Scalability is a critical requirement for data collection mechanisms in large-scale wireless sensor networks, where both the number of sensor nodes and the number of cluster heads may increase significantly. This section discusses how the proposed elastic clustering framework behaves under increasing network scale, in comparison with representative baseline approaches.

The simulation results demonstrate that elastic clustering maintains stable performance as the network size grows from hundreds to thousands of nodes. In particular, network lifetime and control overhead exhibit relatively smooth degradation trends, indicating that the framework adapts effectively to increasing network complexity. This behavior contrasts sharply with baseline schemes, whose performance deteriorates rapidly as network scale increases.

The key factor underlying this difference is the cost of network reconfiguration. In conventional clustering-based and tree-based approaches, increasing the number of nodes or cluster heads amplifies the cost of topology reconstruction. Each cluster head rotation typically triggers global or semi-global control message exchange, leading to control overhead that grows disproportionately with network size. As a result, a large fraction of the network energy budget is consumed by coordination rather than data transmission, causing early node failures and reduced network lifetime.

In contrast, the proposed elastic clustering framework limits the scope of reconfiguration by reusing existing routing and communication structures. When the number of nodes increases, most nodes remain unaffected by cluster head changes and continue to operate using previously established routing paths. Similarly, increasing the number of cluster heads does not lead to proportional growth in control overhead, as topology updates are confined to localized regions rather than propagated across the entire network.

This localized adaptation mechanism enables the framework to exhibit elastic behavior: the network structure stretches and adapts to changes in scale without undergoing disruptive global reconstruction. Consequently, the framework preserves energy efficiency and load balance even in dense and large-scale deployments. These properties are particularly important in raw data collection scenarios, where traffic volume remains high and the margin for inefficiency is limited.

Overall, the scalability analysis confirms that elastic clustering provides a robust and scalable solution for data collection in large-scale wireless sensor networks. By decoupling cluster head rotation from full topology reconstruction, the framework effectively mitigates the scalability *bottlenecks* that constrain conventional approaches.

Chapter Summary

This chapter has presented the implementation and quantitative evaluation of the proposed elastic clustering framework using the OMNeT++ simulator with the Castalia framework. The implementation focused on realizing the core mechanisms introduced in Chapter 3, including dynamic cluster head rotation and topology reuse, within a realistic and scalable simulation environment.

Through extensive simulation experiments, the proposed framework was shown to significantly improve network lifetime and reduce control overhead compared to representative baseline schemes, particularly in large-scale deployments. The results demonstrate that topology reuse plays a central role in mitigating energy

imbalance and limiting the cost of network reconfiguration.

Furthermore, the scalability discussion highlighted the elastic behavior of the proposed framework, showing that its performance degrades gracefully as network size and cluster head count increase, in contrast to conventional approaches that suffer from rapidly increasing coordination overhead.

These findings confirm the effectiveness of elastic clustering as a scalable data collection strategy for large-scale wireless sensor networks. Building on these validated algorithmic contributions, the next chapter introduces a modular wireless sensor network simulation framework built on top of the ns-3 simulator, which serves as a foundation for future system-level extensions and long-term research.

CHAPTER 5. NS-3 BASED WSN FRAMEWORK DESIGN AND CONTRIBUTION

This chapter presents the design and implementation of a wireless sensor network (WSN) simulation framework built on top of the ns-3 simulator. Unlike the previous chapter, which focused on the quantitative evaluation of the proposed elastic clustering framework using the OMNeT++/Castalia environment, the primary objective of this chapter is to introduce and analyze a system-level contribution that supports long-term and extensible research on large-scale wireless sensor networks.

The motivation for developing an ns-3 based WSN framework stems from the evolving requirements of contemporary WSN research. While domain-specific simulators such as Castalia provide convenient abstractions for studying clustering and energy-aware algorithms, their limited extensibility and declining maintenance restrict their suitability for future research directions. In contrast, ns-3 offers high-fidelity protocol implementations, active community support, and strong extensibility toward modern networking paradigms. However, ns-3 lacks native *abstractions* tailored to WSN characteristics, particularly for non-IP, data-centric communication models and large-scale, configuration-driven experimentation.

To address this gap, this chapter introduces a modular *WSN-oriented framework* that operates as an abstraction layer on top of ns-3. The proposed framework aims to combine the realism and sustainability of ns-3 with the usability and configurability traditionally associated with WSN-specific simulators. Rather than serving as a platform for immediate quantitative comparison, the framework is designed to facilitate systematic experimentation, rapid prototyping, and future extension of data collection mechanisms in large-scale WSN scenarios.

The chapter focuses on the architectural design principles, core components, and implementation mechanisms of the framework, including configuration-driven modeling, logical object abstraction, and support for non-IP routing. Through this contribution, the framework establishes a reusable and extensible foundation for subsequent research on advanced routing strategies, dynamic network topologies, and cross-layer optimization in wireless sensor networks.

The remainder of this chapter is organized as follows. Section 5.1 discusses the limitations of existing simulation tools that motivate the need for a new WSN-oriented framework. Section 5.2 presents the design goals and overall architectural approach adopted in this work. Section 5.3 details the configuration-driven

modeling methodology and logical object model of the framework. Section 5.4 introduces the Forwarder abstraction that enables *non-IP* routing support in ns-3. Section 5.5 validates the framework through representative end-to-end simulation scenarios, and Section 5.6 discusses the research value and extensibility of the proposed framework.

5.1 Motivation

This section presents the motivation for developing a wireless sensor network simulation framework based on the ns-3 simulator. The motivation arises from limitations observed in existing WSN simulation tools and from the evolving requirements of contemporary and future research on large-scale wireless sensor networks. These factors highlight a clear gap between algorithm-oriented simulation environments and system-level, extensible research platforms.

5.1.1 Limitations of Existing Simulation Tools

Domain-specific simulation frameworks such as Castalia, built on top of OM-NeT++, have played a significant role in wireless sensor network research for many years. Castalia provides abstractions tailored to WSN characteristics, including sensor node models, energy consumption modeling, clustering-oriented communication, and configuration-driven scenario specification. These features make it particularly suitable for studying algorithmic logic related to clustering, routing, and energy efficiency, as demonstrated in the evaluation presented in Chapter 4.

However, Castalia is no longer actively maintained, and its protocol models are largely abstract representations rather than close implementations of real-world networking stacks. As a result, extending Castalia to support emerging research directions—such as dynamic network topologies, integration with Internet of Things (IoT) systems, or interaction with learning-based control mechanisms—has become increasingly difficult. Moreover, the limited evolution of its codebase restricts long-term sustainability and reuse for future research.

In contrast, ns-3 is an actively developed, general-purpose network simulator with realistic protocol implementations, extensive community support, and strong extensibility. It provides detailed models for wireless communication, mobility, energy sources, and modern networking technologies. Despite these advantages, ns-3 is not inherently designed for wireless sensor networks. It lacks high-level abstractions for WSN-specific concepts such as data-centric communication, non-IP routing, and large-scale configuration-driven experimentation. Consequently, constructing WSN simulation scenarios in ns-3 often requires substantial manual coding, leading to reduced flexibility, lower research productivity, and limited repro-

ducibility.

These observations indicate that existing tools either offer strong WSN abstractions with limited extensibility, or high realism with insufficient support for WSN-oriented modeling. Neither category alone fully satisfies the requirements of scalable and forward-looking WSN research.

5.1.2 Framework Motivation

The framework proposed in this thesis is motivated by the need to bridge the gap between WSN-specific usability and general-purpose simulation realism. Building upon the limitations identified above and the elastic clustering design principles discussed in Section 3.3.2, the objective is to provide a reusable simulation environment that supports both algorithmic exploration and system-level extensibility.

The core motivation is to combine the configuration-driven workflow and hierarchical modeling style of Castalia with the realism, extensibility, and long-term viability of ns-3. By introducing a WSN-oriented abstraction layer on top of ns-3, the proposed framework enables researchers to describe large-scale sensor network scenarios using high-level configuration files, while relying on native ns-3 components for simulation execution. This approach significantly reduces the complexity of scenario construction and lowers the barrier to experimentation in large-scale deployments.

In addition, the framework is designed with future research directions in mind. As WSN research increasingly intersects with mobile data collection, large-scale IoT sensing, and AI-based adaptive networking, a flexible and extensible simulation platform becomes essential. The proposed framework aims to serve as a foundation for such studies by supporting modular component integration, non-IP communication models, and systematic experimentation without requiring invasive modifications to the underlying simulator.

In summary, the motivation for the ns-3 based WSN framework lies not in replacing existing algorithm evaluation platforms, but in complementing them with a system-oriented research tool. This tool is intended to support scalable experimentation, architectural exploration, and long-term research on wireless sensor networks beyond the scope of immediate performance comparison.

5.2 Design Goals

Based on the motivation discussed in the previous section, this section outlines the key design goals that guide the development of the proposed ns-3 based wireless sensor network simulation framework. These goals are formulated to address

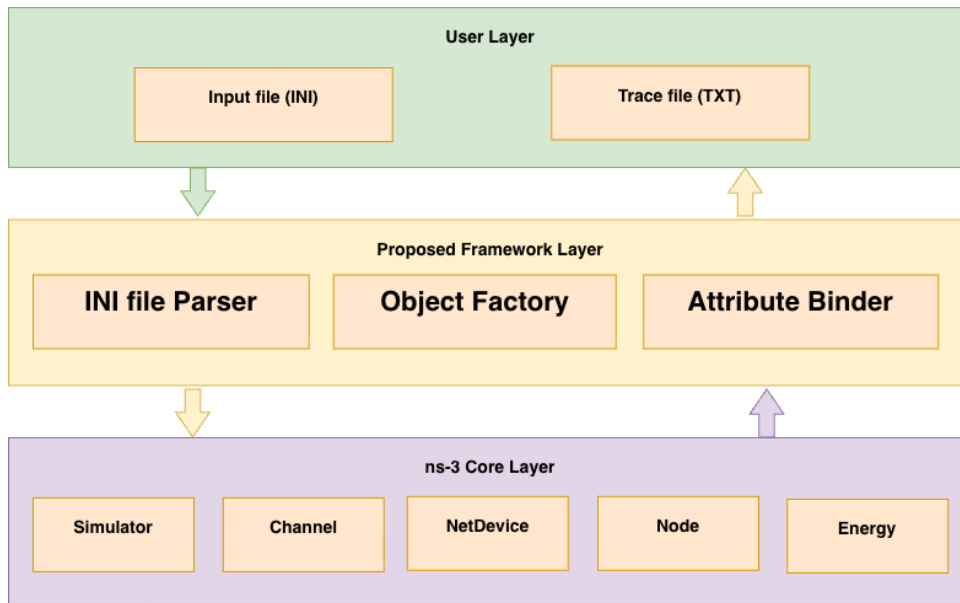


Figure 5.1: NS-3 WSN Framework Architecture Overview

the limitations of existing tools while supporting scalable, extensible, and reproducible research on large-scale WSNs.

The design goals are not tied to a specific data collection algorithm. Instead, they define a general-purpose framework that can accommodate a wide range of WSN research problems, including but not limited to clustering-based data collection, dynamic network adaptation, and future system-level extensions.

Logical Architecture Overview

Figure 5.1 illustrates the logical architecture of the proposed ns-3 based WSN framework. The architecture is organized into three conceptual layers: the *User Layer*, the *Framework Layer*, and the *ns-3 Core Layer*.

The User Layer provides configuration files and experiment scripts that describe network scenarios, protocol composition, and parameter settings without requiring direct modification of source code. The Framework Layer serves as an abstraction layer that interprets configuration descriptions, instantiates logical WSN objects, and orchestrates their interaction. The ns-3 Core Layer executes the simulation using native ns-3 components, including wireless channels, devices, energy models, and event scheduling.

This layered design decouples user-level experimentation from simulator internals, enabling scalable, reproducible, and extensible WSN research.

5.2.1 Configuration-Driven Scenario Specification

A primary design goal of the framework is to support configuration-driven modeling of wireless sensor network scenarios. Rather than requiring researchers to modify source code for each experimental setup, the framework enables network topology, protocol composition, and parameter settings to be described through hierarchical configuration files.

This design is inspired by the productivity advantages observed in Castalia, where large-scale WSN scenarios can be expressed concisely using structured configuration entries and wildcard-based parameter assignment. By adopting a similar approach on top of ns-3, the proposed framework significantly reduces development effort, improves experiment reproducibility, and lowers the barrier to exploring complex or large-scale scenarios.

Configuration-driven specification is particularly important for large-scale WSN research, where small changes in network size, topology, or protocol parameters may require extensive code modifications in traditional ns-3 simulations. By externalizing these aspects into configuration files, the framework promotes rapid experimentation and systematic performance analysis.

5.2.2 Logical Object Model for WSN Abstraction

Another key design goal is to introduce a logical object model that explicitly represents the natural structure of wireless sensor networks. The framework defines a set of logical entities, such as *SensorNetwork*, *Node*, *Routing*, *MAC*, *Radio*, and *Application*, which reflect how WSNs are commonly conceptualized at the system level.

These logical objects serve as an intermediate abstraction layer between high-level configuration descriptions and the concrete runtime components provided by ns-3. By decoupling logical modeling from simulator-specific implementation details, the framework enables clearer system architecture, improved modularity, and easier reasoning about network behavior.

This object model also facilitates systematic extension of the framework. New protocols, energy models, or application behaviors can be introduced by extending the logical object hierarchy without altering the core framework structure. As a result, the framework supports both algorithm-oriented experimentation and architectural exploration.

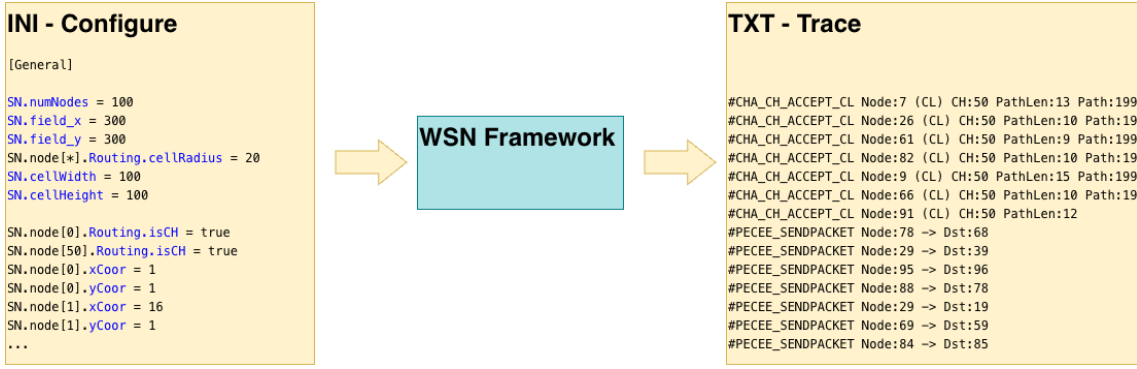


Figure 5.2: Example of Framework Usage

5.2.3 Factory and Registry-Based Extensibility

To ensure flexibility and long-term extensibility, the framework is designed around a factory and registry mechanism for object creation and management. Rather than hard-coding component types and dependencies, the framework dynamically instantiates logical objects based on type identifiers specified in configuration files.

The factory mechanism maps abstract component identifiers to concrete implementations, while the registry resolves hierarchical relationships between objects and manages their lifecycle. This design allows new components—such as routing protocols, MAC layers, or energy models—to be added without modifying existing framework code.

By isolating extension points and avoiding tight coupling between components, the framework supports incremental development and experimentation. This property is essential for WSN research, where new algorithms and system designs are frequently introduced and evaluated under varying assumptions.

Framework Initialization Workflow

Algorithm 3 summarizes the high-level workflow of building and initializing a WSN simulation scenario using the proposed ns-3 based framework. The pseudo-code emphasizes configuration-driven construction and separation between logical modeling and runtime execution.

5.2.4 Support for Non-IP and Data-Centric Communication

A central design goal of the proposed framework is to support communication models that are intrinsic to wireless sensor networks but poorly represented in traditional IP-centric simulation environments. Many WSN protocols rely on hop-by-hop forwarding, data-centric routing, and cross-layer interaction, which do not naturally map onto IP-based abstractions.

Algorithm 3 Framework Build and Initialization Workflow

Input: Hierarchical configuration files
Output: Executable ns-3 simulation scenario

- 1: Parse configuration files
- 2: Instantiate *SensorNetwork* object
- 3: **for all** node definitions in configuration **do**
- 4: Create logical *Node* object
- 5: Register node in *SensorNetwork*
- 6: **end for**
- 7: **for all** protocol components (Routing, MAC, Radio, Application) **do**
- 8: Use Factory to instantiate component
- 9: Bind component to corresponding logical Node
- 10: **end for**
- 11: Resolve dependencies via Registry
- 12: Map logical objects to ns-3 runtime components
- 13: Initialize energy models and communication devices
- 14: Start simulation execution in ns-3 core

To address this gap, the framework introduces a *Forwarder* abstraction that functions as a lightweight network-layer component independent of IP. The Forwarder enables explicit control over packet forwarding logic, allowing the implementation of routing schemes that operate directly on data attributes or local network state.

This abstraction is critical for realizing realistic WSN protocols in ns-3 without forcing them into unsuitable IP-based models. It also provides a foundation for advanced research topics such as adaptive routing, cross-layer optimization, and learning-based control mechanisms.

Forwarder-Based Non-IP Routing Logic

Algorithm 4 illustrates the abstract forwarding logic enabled by the Forwarder component. The pseudo-code highlights how packet forwarding decisions are made without relying on IP addressing or traditional network-layer protocols.

5.2.5 Scalability and Research-Oriented Design

Finally, the framework is designed with scalability and long-term research value as primary considerations. The architectural separation between configuration, logical modeling, and runtime execution ensures that large-scale scenarios can be constructed and simulated without excessive complexity.

Rather than optimizing for a single evaluation study, the framework aims to serve as a reusable research platform that can evolve alongside emerging WSN research directions. This includes support for large-scale IoT sensing, mobile data

Algorithm 4 Forwarder Abstraction for Non-IP Routing

Upon receiving packet p from lower layer

- 1: Extract metadata from p ▷ e.g., data type, cell ID, role
- 2: Consult local routing state
- 3: $nextHop \leftarrow \text{SelectForwardingTarget}(p)$
- 4: **if** $nextHop$ exists **then**
- 5: Forward p to $nextHop$
- 6: **else**
- 7: Drop or buffer p
- 8: **end if**

Upon local state update

- 9: Update forwarding rules accordingly

collection, and integration with intelligent decision-making mechanisms.

Together, these design goals define a framework that complements algorithm-focused evaluation tools by providing a flexible, extensible, and system-oriented simulation environment. The following sections describe how these goals are realized through concrete architectural choices and implementation mechanisms.

5.3 Research Value and Extensibility

This section discusses the research value of the proposed ns-3 based wireless sensor network framework and its potential for future extensions. Unlike the quantitative evaluation presented in Chapter 4, the focus here is on validating the feasibility of the framework as a research platform and highlighting its long-term applicability to a broad range of WSN-related studies.

5.3.1 Framework Validation

The primary objective of the ns-3 implementation presented in this chapter is to validate that the proposed framework can operate correctly in an end-to-end simulation workflow. Rather than serving as a benchmark for algorithmic performance comparison, the validation aims to demonstrate that the framework architecture, abstraction layers, and configuration mechanisms function cohesively within the ns-3 simulation environment.

Basic validation scenarios are constructed to verify the following aspects. First, the framework is able to parse hierarchical configuration files and construct a corresponding logical object model that accurately represents the intended wireless sensor network structure. Second, logical objects are correctly instantiated and bound to ns-3 runtime components during the build phase, ensuring consistency between configuration-level descriptions and simulation-level execution. Third, data packets can be generated at sensor nodes, forwarded hop-by-hop through the

Table 5.1: Comparison of simulation execution time per cluster head rotation

Network Size	Castalia	ns-3 Framework	Relative Speedup
100	~8.11 s	~ 2.20 s	$\approx 3.7\times$
400	~19.67 s	~ 4.8 s	$\approx 4.1\times$
1024	~204.8 s	~42 s	$\approx 4.8\times$

Forwarder abstraction, and delivered to designated sinks without relying on IP-based communication.

These validation scenarios confirm that the framework supports complete simulation execution from scenario specification to result generation. They also demonstrate that the introduced abstractions, such as the logical object model and non-IP forwarding mechanism, integrate correctly with ns-3’s event-driven simulation core. This validation establishes confidence that the framework is suitable for implementing and experimenting with WSN protocols at the system level.

5.3.2 Simulation Performance and Scalability Benefits

Beyond architectural extensibility and modeling flexibility, the transition from Castalia to an ns-3 based simulation framework also provides tangible improvements in simulation performance, particularly for large-scale wireless sensor network scenarios. As network size increases, the execution time of simulation experiments becomes a practical limiting factor for systematic experimentation, especially when repeated cluster head rotation and multi-round evaluation are required.

In existing Castalia-based implementations, simulation runtime grows rapidly with network size due to increasing event density, centralized control logic, and limited optimization for large-scale execution. These limitations become especially evident in networks consisting of hundreds to thousands of nodes, where the time required to simulate a single data collection round or cluster head rotation can significantly slow down experimental workflows.

By contrast, the ns-3 simulation core provides a more efficient event scheduler, optimized memory management, and improved scalability for large-scale network execution. When combined with the lightweight, non-IP Forwarder abstraction introduced in this framework, ns-3 enables substantially faster simulation of large WSN deployments while preserving protocol fidelity and execution correctness.

The results in Table 5.1 illustrate that the performance advantage of the ns-3 based framework becomes increasingly pronounced as network size grows. While the difference in execution time is modest for small-scale scenarios, ns-3 exhibits significantly better scalability for networks of several hundred to over one thousand

nodes.

This improvement is particularly important for studies involving frequent cluster head rotation and multi-round evaluation, where total simulation time can otherwise become prohibitive. Faster execution not only improves research productivity but also enables more extensive parameter exploration and sensitivity analysis in large-scale WSN experiments.

These results confirm that the ns-3 based framework contributes not only architectural extensibility and long-term sustainability, but also practical performance benefits that are critical for large-scale simulation-based research.

It is important to note that the simulation runtime results reported above should be interpreted as *preliminary and scenario-specific*. The presented comparison reflects the execution performance of the proposed ns-3 based framework under a simplified PECEE configuration, which represents the most basic form of elastic clustering without advanced coordination, learning-based decision making, or dynamic topology adaptation.

At the current stage of the research, the PECEE implementation focuses on validating the feasibility of large-scale simulation, topology reuse, and localized reconfiguration. Consequently, the computational complexity of node behavior, control logic, and inter-layer interaction remains relatively limited. Under these conditions, the efficiency of the ns-3 simulation core and the lightweight Forwarder abstraction lead to noticeable runtime improvements compared to Castalia.

As the research progresses, future extensions of the framework are expected to incorporate more complex system components, including heterogeneous IoT devices, intelligent control mechanisms based on machine learning or artificial intelligence, and mobility support for nodes or data collectors. Such extensions will inevitably increase the computational complexity of both protocol logic and simulation execution, potentially reducing or even eliminating the observed runtime advantage.

Therefore, the results reported in Table 5.1 should not be interpreted as a general claim of superior simulation performance under all conditions, but rather as an initial indication of the framework's ability to efficiently support large-scale WSN experimentation in its current form. This clarification reinforces the intended role of the ns-3 based framework as a flexible and extensible research platform, whose performance characteristics evolve together with the complexity of the modeled system.

5.3.3 Research Value and Extensibility

Beyond functional validation, the proposed framework offers significant research value due to its architectural design choices. By combining layered abstraction, configuration-driven modeling, and non-IP communication support, the framework provides a flexible foundation for exploring a wide range of wireless sensor network research problems.

The separation between high-level configuration, logical modeling, and runtime execution enables researchers to rapidly prototype new protocols or system architectures without modifying the simulator core. This property is particularly valuable for studies on adaptive routing, where protocol behavior may change dynamically in response to network conditions, energy state, or application requirements.

The Forwarder abstraction further extends the applicability of the framework by enabling data-centric and cross-layer routing mechanisms that are difficult to express using IP-centric models. This makes the framework well suited for investigating emerging WSN paradigms, including delay-tolerant data collection, opportunistic forwarding, and in-network decision making.

In addition, the framework is designed to accommodate dynamic and heterogeneous network scenarios. Extensions toward mobile sinks, UAV-assisted data collection, and hybrid static–mobile sensor networks can be naturally supported by integrating mobility models and control logic at the framework level, while continuing to rely on ns-3 for realistic communication modeling.

Finally, the modular architecture and clear extension points make the framework a promising platform for future research involving intelligent network control. The logical object model and configuration-driven design provide natural integration points for AI-, machine learning-, or reinforcement learning-based adaptation mechanisms. Such approaches may be used to optimize cluster head selection, routing decisions, or resource allocation in large-scale wireless sensor networks.

Overall, the proposed ns-3 based framework serves not only as a validation of the elastic clustering concept in a general-purpose simulator, but also as a reusable and extensible research platform. Its design supports long-term experimentation and system-level exploration, complementing the algorithm-focused evaluation presented earlier in this thesis and enabling future advances in large-scale wireless sensor network research.

Chapter Summary

This chapter presented the design and implementation of a wireless sensor network simulation framework built on top of the ns-3 simulator, with the objective of supporting long-term and extensible research on large-scale WSN data collection. Unlike the performance-oriented evaluation conducted in Chapter 4, the emphasis of this chapter was placed on system architecture, abstraction design, and research applicability rather than quantitative algorithm comparison.

The chapter first analyzed the limitations of existing WSN simulation tools, highlighting the gap between domain-specific frameworks that favor usability and general-purpose simulators that emphasize realism and long-term maintainability. Motivated by this gap, a configuration-driven and WSN-oriented framework was introduced to bridge the strengths of Castalia-like modeling with the extensibility and protocol realism of ns-3.

The core architectural contributions of the framework include a hierarchical configuration mechanism, a logical object model that reflects the natural structure of wireless sensor networks, and a factory–registry-based instantiation process that decouples scenario description from runtime execution. In addition, the Forwarder abstraction was introduced as a key system-level contribution, enabling non-IP, hop-by-hop communication and data-centric routing models that are difficult to express using conventional IP-based approaches in ns-3.

Through end-to-end validation scenarios, the framework was shown to operate coherently within the ns-3 simulation environment, confirming the feasibility of realizing the proposed elastic clustering design in a general-purpose network simulator. More importantly, the framework establishes a reusable and extensible research platform that supports future studies on adaptive routing, dynamic network structures, UAV-assisted data collection, and intelligent network control mechanisms.

By complementing the algorithmic evaluation presented earlier in this thesis, this chapter extends the contribution of the work from a specific data collection solution to a broader system-level research foundation. The next chapter concludes the thesis by summarizing the main findings and contributions, and by outlining promising directions for future research that build upon both the elastic clustering framework and the ns-3 based simulation platform introduced in this work.

CHAPTER 6. CONCLUSIONS

This thesis investigated the problem of efficient data collection in large-scale wireless sensor networks under stringent energy constraints and raw data transmission requirements. Motivated by the scalability limitations of existing clustering-based and routing-based approaches, particularly the high control overhead induced by frequent cluster head rotation, this work proposed an elastic clustering framework that emphasizes topology reuse and localized network reconfiguration.

The core contribution of the proposed framework lies in decoupling high-level clustering decisions from low-level topology management. By preserving and reusing previously established local routing structures, elastic clustering enables dynamic cluster head rotation without triggering network-wide topology reconstruction. This design significantly reduces control overhead, mitigates energy imbalance, and supports stable long-term operation as network scale increases.

The effectiveness of the proposed approach was validated through extensive simulation-based evaluation using the OMNeT++/Castalia framework. Quantitative results demonstrated that elastic clustering achieves longer network lifetime, improved energy balance, and substantially lower cumulative control overhead compared with representative baseline schemes, including GSTEB and HTC-RDC, especially in large-scale and raw data collection scenarios.

In addition to algorithmic contributions, this thesis introduced a modular wireless sensor network simulation framework built on top of the ns-3 simulator. While not intended for direct performance comparison, the ns-3 based framework validates the feasibility of realizing elastic clustering in a general-purpose simulation environment and establishes a reusable and extensible foundation for future system-level research on large-scale wireless sensor networks.

Overall, the results confirm that elastic clustering combined with topology reuse provides an effective and scalable solution for energy-efficient data collection in large-scale wireless sensor networks. By addressing both algorithmic and system-level challenges, this work contributes practical insights and research infrastructure to support future advances in wireless sensor network design.

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APPENDIX

A. ACCEPTED PUBLICATION

This appendix includes the full accepted manuscript that corresponds to the core contributions of this thesis.

The paper has been accepted for publication at an international peer-reviewed IEEE conference. The manuscript is reproduced here verbatim for completeness.

An Elastic Clustering Framework for Large-Scale WSNs Maximizing Network Lifetime

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Abstract—Large-scale Wireless Sensor Networks (WSNs) are applied in multiple domains, notably Precision Agriculture, Intelligent Traffic Management and Smart Hospital. A key task for these applications, long-term environment monitoring, relies on periodic data collection via network clustering. In applications like precision agriculture, large networks requires frequent topology updates to avoid hotspot issues at Cluster Heads (CHs), balance energy consumption, and meet QoS demands. Centralized topology maintenance at the Base Station ensures global energy efficiency but incurs costly network-wide control overhead in larger deployments. This undermines existing LEACH-derived and even recent protocols using tree-based approaches, in terms of scalability. We propose PECEE, a framework combining centralized CH selection with local routing coordination, enabling elastic clustering for raw data collection. Simulations show that PECEE extends network lifetime from 50% to 90% compared to some notable solutions including GSTEB in raw data collection simulations, and up to 200% when data aggregation is used.

I. INTRODUCTION

Large-scale wireless sensor networks(WSNs) have been widely deployed worldwide, delivering remarkable results in various practical applications. Notable examples include precision agriculture and intelligent transportation management. For instance, the DEMETER project has deployed over 38000 devices including sensor for agricultural monitoring across 18 countries in 25 pilot sites in Europe [1]. Similarly, the Caltrans Performance Measurement System(PeMS) by the California Department of Transportation(USA) provides real-time traffic monitoring on major highways and urban areas with the use of nearly 40,000 individual detectors across California [2]. In these large-scale applications, the central base station not only collects locally aggregated or processed data but also substantial volumes of raw data, such as images of crop diseases or traffic incidents.

Over the past two to three decades, the problem of continuous and periodic data harvesting in WSNs has garnered significant attention from the research community, with typical solution approaches relying on clustering-based processing. However, traditional protocols such as LEACH and PEGASIS are ill-suited for large-scale networks(typically limited to around 100 nodes or fewer), as they fail to provide efficient solutions for data transmission from distant clusters to the base station. Recent solutions for large networks still adhere to clustering paradigms but require deep data aggregation

from clusters(and even lower tiers) to minimize the volume transmitted to the base station. The most critical challenge in large-scale networks is the rapid energy depletion of hotspot nodes (those near the base station or cluster heads), where traffic is concentrated, severely impacting network lifetime.

Recently, solutions leveraging mobile sinks, including unmanned aerial vehicles (UAVs or drones), have attracted considerable interest. For example, in precision agriculture, UAVs can be dispatched to continuously fly to cluster heads (CHs) to directly collect data via wireless channels before returning to the base station for offloading. This approach completely alleviates the network transmission burden from CHs to the base station. To ensure load balancing and prevent energy exhaustion at CHs, cluster head rotation techniques are commonly employed. Nevertheless, prior studies suffer from a major limitation: CH rotation incurs substantial overhead due to the complete reconstruction of network topology at each round, thereby negatively affecting network lifetime.

Typically, proposed schemes focus on selecting new CHs for each round using intelligent heuristics that prioritize healthier nodes distributed evenly across the network. However, this is followed by rebuilding clusters and internal data collection structures from scratch in each cluster, resulting in considerable communication overhead. Thus, for data harvesting tasks in clustering-based WSNs, a core challenge persists: how to enable CH rotation while avoiding excessive overhead. In other words, after generating a list of CH changes (additions and removals), the system must efficiently re-establish network linkages to minimize overhead comprehensively.

To address this core challenge, we designed an efficient solution that provides a platform for the continuous reconfiguration of clusters and their internal connection structures whenever CH rotation is required. This platform offers services to the upper layer, which submits reconfiguration requests along with a new CH list (generatable by any pre-selected rotation algorithm). The key innovation of our solution lies in maximizing the inheritance and reuse of existing network connection components from the previous round. Our approach is highly generalizable, serving as a platform for most prior important schemes, which primarily focus on new CH placement, thereby creating the most efficient overall system through optimal integration of infrastructure and upper-layer components.

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We propose PECEE¹, a novel framework for optimizing cluster reconfiguration and energy efficiency in large-scale WSNs. Simulations show that PECEE extends network lifetime from 50% to 90% compared to some notable solutions, including GSTEB.

II. RELATED WORKS

Research on Wireless Sensor Networks (WSNs) has centered on two technical pillars: *Cluster Head (CH) selection* and *routing*. These determine network lifetime and scalability, the main concerns for applications such as precision agriculture, where thousands of nodes must transmit raw data.

A. Cluster-Head Selection

LEACH established clustering as the foundation for energy-efficient communication through distributed CH rotation [3]. It required no central control but often ran into the problem of selecting low energy CHs. Later, HEED [4] and DEEC [5] incorporated residual energy and node degree for energy-aware CH election while adopting the distributed execution mode.

Centralized CH selection emerged with LEACH-C [6], where the Base Station (BS) used global energy and node location data to optimize CH placement. This improved energy distribution and network lifetime but incurred control overhead. Metaheuristic nature-inspired optimizations further refined CH placement for heterogeneous or larger networks [7].

Recent work integrates CH selection with application-level objectives. In UAV-assisted WSNs, for example, CH placement is jointly optimized with UAV trajectories to balance load and extend lifetime [8]. While efficient, these centralized schemes incur high reconfiguration costs whenever CHs change.

B. Routing in Clustered Networks

Routing governs data flow between nodes, CHs, and the BS. In LEACH, routing follows the star topology, nodes transmit directly to the CH, which aggregates and forwards data directly to the BS. This quickly becomes inefficient when clusters size scaled due to long single-hop links. PEGASIS [9] replaced this with a chain structure to reduce long-distance links, while GSTEB [11] introduced a tree-based model to distribute forwarding load and enhance scalability. These chain and tree structures remain key alternatives for intra-cluster routing.

In scaled networks where CH can not reach the BS in a direct manner, hierarchical routing was introduced. MH-LEACH [10] added multi-hop paths among CHs, improving network scalability and balancing energy across CHs. Cluster-based Routing Protocol (CRP) family expanded on this multi-hop idea for large-scale networks [12]. Later UAV-aided WSN generation alleviated the problem of inter-cluster routing [8].

However, *most clustered routing protocols rely on data aggregation to limit transmissions*, unsuitable for applications

requiring high-fidelity raw data collection such as precision agriculture or multimedia sensing. They demand routing schemes that preserve data fidelity while remaining energy-efficient. GSTEB proposed an algorithm for these applications [11]. Recently, HTC-RDC revisited this demand and proposed a load-balanced tree-construction strategy [13].

C. Emergence of the Elastic Cluster

Centralized CH selection offers effective global energy management, but frequent CH reassignment each round incurs costly network-wide updates from the BS, particularly in large scale networks. Existing approaches are insufficient for large, long-lived networks that support raw data collection. This paper tackles the challenge of combining the efficiency of centralized CH selection with the flexibility of localized routing to achieve *Elastic Clustering*: adaptive clusters formation and routing for raw data collection, while minimizing coordination overhead and extending network lifetime.

III. PECEE: PLATFORM FOR ELASTIC CLUSTERING AND ENERGY EFFICIENCY

This section describe our system model, the main idea of PECEE, and then elaborate on the key issues and technical details

A. System Model

Wireless transmissions: Wireless Sensor Nodes utilize the radio spectrum to transmit and receive data packets. We assume the unit disk transmission model in this paper, with an uniform *transmissionRange* amongst all Nodes.

Network Model: The size of large scale monitoring WSNs can span between hundreds to thousands of Sensor Nodes, deployed in a grid-like manner to ensure the sensing coverage.

We assume several key characteristics of this network:

- **Sparse Spacing**: Nodes are sparsely spaced at approximately their *transmissionRange*
- **Raw Data Collection**: Data packets travelling from all of the Nodes, up the CFT is not modified, each Parent Node simply relays its children's packets.
- **Node Homogeneity**: All Nodes have the same hardware specification: energy amount, radio transmission range and computation power.

B. Main Idea

The core idea of PECEE is to conduct an *initial network setup* phase prior to data harvesting, establishing a cellular network composed of autonomous cells, each with a topology designed for long-term utilization. Concurrently, a cell coloring mechanism is implemented to ensure that cells sharing the same color (i.e., communicating simultaneously) are free from mutual radio interference. During this phase, "pending links" between adjacent cells are established and configured for reuse in subsequent operations as required.

The data collection process is then organized into rounds, each comprising two phases: the *Setup Phase* and the *Steady Phase*, inspired by the LEACH approach. During the Setup

¹Platform for Elastic Clustering and Energy Efficiency

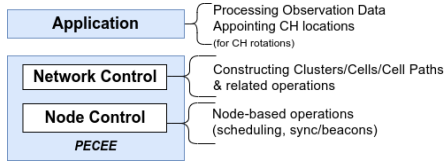


Fig. 1. Architecture of our WSN system.

Phase, clusters are formed by grouping cells based on their self-selected affiliations. *Cell Paths* are generated to define straightforward data transmission routes toward the *Cluster Head(CH)*, leveraging pre-established “pending links” at the boundaries of neighboring cells. In the Steady Phase, operating cyclically based on color schedules, cells of a given color forward data toward their respective CHs along the predetermined *Cell Paths*.

From a system perspective, the network is organized into three layers. The highest layer, the Application, hosts a service that enables users to utilize the network. This layer defines mechanisms for selecting CHs during each round(CH rotation) and application-specific criteria and Quality of Service(QoS) requirements (e.g., criteria for prioritizing which cluster each node should join), which will be discussed further in Section 4. The second layer, Network Control, provides mechanisms for establishing the cell grid and forming clusters. The lowest layer, Node Control, manages direct communication between neighboring nodes, scheduling transmissions to avoid interference.

C. The Cellular Overlay

PECEE organizes the network as a set of autonomous cells based on a regular hexagonal grid. Using GPS, nodes determine their coordinates and identify their corresponding cell within the hexagonal grid. A node closest to the cell’s center is designated as the *Cell Leader(CL)*. The topology connecting nodes within a cell is constructed by the CL, utilizing node updates on their positions and neighbors(nodes communicate with neighbors via a standard beacon mechanism). The establishment of the cell grid, CLs, and local cell topologies occurs during the initial network setup phase. Simultaneously, the CL designates nodes at the cell boundaries as *Cell Gateways(CGs)*, which serve as connection points for data transmission to neighboring cells. These CGs act as “pending links” to establish cell paths connecting to Cluster Heads(CHs) in the future. With each CG as the root, a *Cell Forwarding Tree(CFT)* and transmission schedule are established, enabling other nodes in the cell to forward data to the CG when needed.

Due to page constraints, we refrain from elaborating on the detailed procedures for the network setup; comprehensive algorithmic pseudocode will be provided in the extended version of this paper. Instead, illustrative figures(Fig. 2, Fig. 4, and Fig. 6) are included to elucidate the nature and significance of these steps.

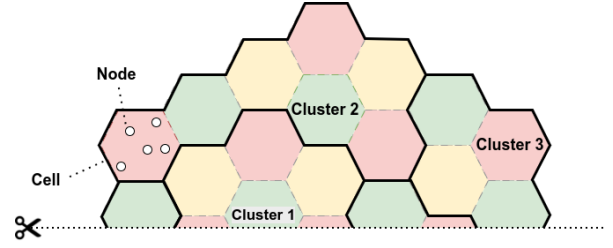


Fig. 2. The Cellular Overlay

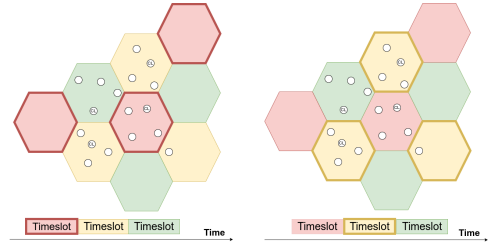


Fig. 3. Example of constraining transmissions with the Cellular Overlay over the time axis(TDMA)

To prevent mutual radio interference between non-adjacent cells, the edge length of each hexagonal cell is designed to exceed the transmission radius, with the ratio($\epsilon > 1$) serving as a configurable network parameter. A minimum of three colors is required to color the cell grid, ensuring that no two adjacent cells share the same color. For visualization, we adopt three distinct colors: Red, Green, and Yellow. As shown in Fig. 2, cells are arranged in a hexagonal pattern, with each pair of adjacent cells assigned different colors to avoid interference. Fig. 3 illustrates the transmission schedule, where cells of each color take turns transmitting in designated time slots, ensuring that only nodes in cells of the active color communicate with their neighbors during a given slot.

Fig. 4 illustrates the role of CG nodes: each CG serves as a “pending link” connecting a cell to its adjacent neighbor at the boundary, with a *Cell Forwarding Tree* rooted at the CG to define how nodes within the cell forward data toward the CG for subsequent transmission to the corresponding neighboring cell.

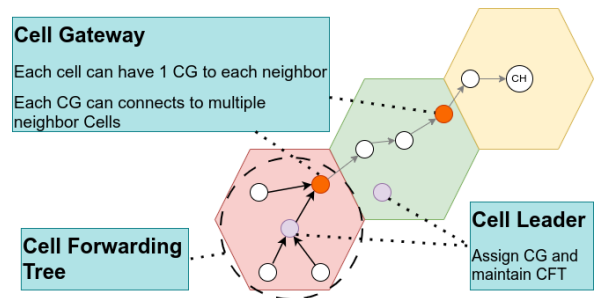


Fig. 4. Cell coordination and routing.

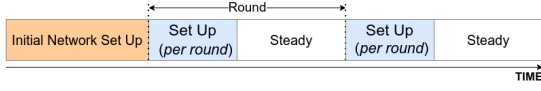


Fig. 5. Main Phases

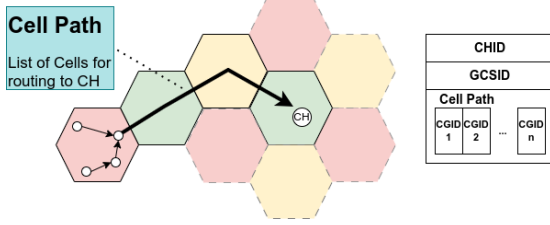


Fig. 6. Cell Path and CHA Packet

D. Main Phases

In the Setup Phase of each round, when the Cluster Head(CH) list is updated through CH rotation, each new CH broadcasts a *Cluster Head Advertisement*(CHA) packet announcing its location to the surrounding area. Each Cell Leader receives multiple CHA packets and autonomously decides which cluster to join, based on policies set by the Application Layer. Upon receiving a CHA packet, a cell forwards it to neighboring cells after updating the *Cell Path*(CP) field in the packet, which includes the IDs of cells traversed. Once a CL determines the cluster it wishes to join, it selects the corresponding CHA to establish a Cell Path, defining the route to the CH. Each cell also identifies the corresponding CGs along the CP. Fig. 6 illustrates how a CP is constructed from the current cell to the newly established CH. These CPs are formed by leveraging the interconnection of necessary CGs along the designated route.

Through the topology setup steps in the Setup Phase, data collection in the Steady Phase becomes straightforward: each cell accumulates data via the CFT(following the associated transmission schedule), and the CG forwards the accumulated data to the next cell along the CP. Below, we provide a more detailed description of the setup steps in the Setup Phase.

When a CL receives multiple CHA packets from various new CHs, it updates its environmental state, including information about existing clusters(identified through old and new CHs it has interacted with), distances, and energy consumption levels. Based on the selection criteria and procedures specified by the Application Layer, CLs choose the most suitable cluster to join and establish corresponding technical parameters. A common criterion is to prioritize distance, where the CL simply selects the CH with the shortest cell-path. However, alternative criteria may be imposed by the Application Layer, such as minimizing the energy cost of transmitting each packet to the CH.

Our proposed PECEE solution serves as a platform that provides flexible algorithmic mechanisms to support various data harvesting applications with diverse objectives and Quality of Service(QoS) requirements. To achieve this, the frame-

work incorporates modular optimization slots that function as callbacks for application-specific algorithms (e.g., for cluster construction or selection of CLs/CGs). These algorithms are referred to as *Global Clustering Strategies*(GCS). The general mechanism for cluster construction is described through algorithmic pseudocode in Algo , where line 9 specifies the invocation of an appropriate GCS based on the application-specific context. The subsequent Section 4 provides specific application examples.

Algorithm Cluster Formation

(executed by each CL)

```

1: State:
2:  $CH_{Aset} \leftarrow$  all received CH Advertisements(CHA)
3: if node is Cluster Head( $CH_i$ ) then
4: Broadcast CHA( $CH_i$ ) to all neighboring Cells
5: Upon receiving CHA( $CH_i$ ) from neighbor  $CL_k$ :
6:  $CH_{Aset} \leftarrow CH_{Aset} \cup \{CHA(CH_i)\}$ 
7: Rebroadcast CHA( $CH_i$ ) to all neighbors except  $CL_k$ 
8: After CHA flood timeout  $T_f$ :
   {Determine best CH according to selection policy}
9:  $CH \leftarrow \mathbf{CHSelect-GCS}(CH_{Aset})$ 
10: if CH in Cell then
11:    $CG \leftarrow CH$ 
12:    $CFT \leftarrow \mathbf{calculateAndCacheCFT}(CG)$ 
13: else
14:    $nextCell \leftarrow$  next Cell to CH(from CHA.CellPath)
   {Reuse cached topologies from Initial Network Setup}
15:    $CG \leftarrow \mathbf{getCachedCG}(nextCell)$ 
16:    $CFT \leftarrow \mathbf{getCachedCFT}(CG)$ 
17: end if
   {Synchronize routing state locally}
18: Update CFT info for all Cell Members
19: Result: Each Cell Leader selects next round CH, Cell Path, CG and CFT.

```

IV. TYPICAL SCENARIOS IN USING PECEE

In this section, we discuss several typical applications where our proposed solution provides a robust platform to support critical real-world use cases.

Precision Agriculture. In this scenario, on-top applications may consist of services that continuously monitor and control the agricultural environment, equipped with a centralized optimization algorithm for CH selection. PECEE accepts requests with a list of CHs as input for each round(computed by the BS using the aforementioned centralized optimization module), then configures the network accordingly and manages the underlying system to collect and forward data to the CHs. The mechanism for relaying data from CHs to the BS, which is pre-selected by the application (e.g., using drones for data collection), is outside the scope of PECEE.

With the provided WSN platform, nodes periodically forward observation data to their respective Cluster Head(CH), which subsequently relays the data to the Base Station(BS) via

hop-to-hop communication or through UAVs/drones. Alongside domain-specific agricultural data, information about the remaining energy levels of nodes is also transmitted. This enables the BS to maintain a comprehensive overview of the network and make optimal decisions for designating CHs in subsequent rounds, replacing those that have operated for extended periods and exhibit significantly reduced energy levels compared to the network average. The designated CHs receive notifications from the BS and broadcast them within their local area, allowing nodes in the region to join an appropriate cluster based on defined selection criteria. These criteria may be application-specific, such as prioritizing the closest geographical distance to the CH or minimizing energy consumption for hop-to-hop communication to the CH. A third, more complex option involves balancing the energy consumption of a CH (dependent on cluster size) proportionally to its remaining energy. While this criterion is more challenging to satisfy, it helps facilitate load balancing and effectively extends network lifetime. To meet these criteria, PECEE incorporates pre-implemented modular algorithms (i.e., GCSs mentioned earlier) within the platform, which are activated based on the requirements of specific applications.

Intelligent Traffic Management. In an intelligent traffic management scenario, vehicles moving to a new region may re-register with a different cell and cluster. To navigate efficiently, each vehicle can connect to the CH to obtain updated intra-cluster traffic information. In the event of a traffic incident, nodes within a cell promptly report updates to the CH. Nodes equipped with cameras can proactively capture footage and send it to the CH for potential forwarding to the BS. Data from the CH can be relayed to Roadside Units (RSUs) for processing by AI systems, leveraging edge computing and fog computing capabilities. For complex traffic incidents generating large streaming data volumes, data can be transmitted via multiple cell paths to RSUs in neighboring clusters, enabling rapid delivery to a central processing unit with sufficient AI computational resources.

Smart Hospital IoT Network. Through the network of cells, CHs continuously collect information about hospital pathways, corridors, crowded patient areas, and emergency incident locations. This enables the system to coordinate and route Automated Guided Vehicles (AGVs) for transporting equipment, supplies, and medications to their destinations according to scheduled timelines.

V. SIMULATION AND EVALUATION

A. Settings

We evaluate the performance of PECEE through simulation-based experiments. Two simulation scenarios are considered: the Raw Data (RD) Scenario and the Data Aggregation (DA) Scenario, with a focus on the former due to its greater challenge in fully relaying data to the Base Station (BS). In the DA Scenario, data received at each intermediate node is aggregated into a single packet for forwarding. In both scenarios, we compare PECEE against a prominent baseline, GSTEB. Additionally, in the RD Scenario, we include HTC-RDC, a

TABLE I
SIMULATION PARAMETERS

Parameter	Setting
Initial Energy	0.1 J per node
Node Density	0.0004 nodes/m ²
Sensing Rate	1 pkt/round
Path Loss Model	Free-space

recent state-of-the-art method supporting RD, for comparison. Since existing solutions for RD often focus on building energy-efficient trees and lack clustering, we augment them with a cluster construction procedure using Voronoi-based spatial clustering method. All simulations employ the same CH rotation algorithm, which selects new Cluster Heads (CHs) per round based on higher residual energy probabilities. Experiments are conducted using the OMNET++ simulation platform, integrated with the Castalia framework [14] for MAC-layer communication. Common simulation parameters are detailed in Table I.

B. Results

1) *Network Lifetime:* We ran simulations on 3 network sizes, to observe the performance of each solution for large-scale networks. Network Lifetime is defined as the round in which the first node depletes its energy (First Node Death).

TABLE II
NETWORK LIFETIME FOR RAW DATA SCENARIO IN ROUNDS

Network Settings		Routing Algorithm		
Size	CHs count	PECEE	GSTEB	HTC-RDC
100	3	340	152	253
400	6	174	61	92
400	12	190	88	163
400	24	195	158	173
900	27	178	125	130

TABLE III
NETWORK LIFETIME FOR DATA AGGREGATION SCENARIO IN ROUNDS

Network Settings		Routing Algorithm	
Size	CHs count	PECEE	GSTEB
100	3	480	357
400	12	901	281

2) *Node Energy over lifetime:* The long-term stability of the network is evaluated using node residual energy. We collect minimum and average residual energy value of nodes at end of each round. See Fig 7

3) *Cumulative Control Overhead:* The Cumulative Control Overhead (CCO) at round i represents the total network-wide energy spent for control transmissions up to that round, computed as $CCO_i = \frac{\sum_{r=1}^i \text{Transmitted Control Bits}_r}{\sum_{r=1}^i \text{Total Transmitted Bits}_r}$. Lower CCO indicates higher energy saving in control operations.

C. Analysis

In smaller networks, PECEE performs comparably to GSTEB and HTC-RDC but excels in larger networks, achieving 30-90% longer lifetime (see Table II, Fig 7). Table II also

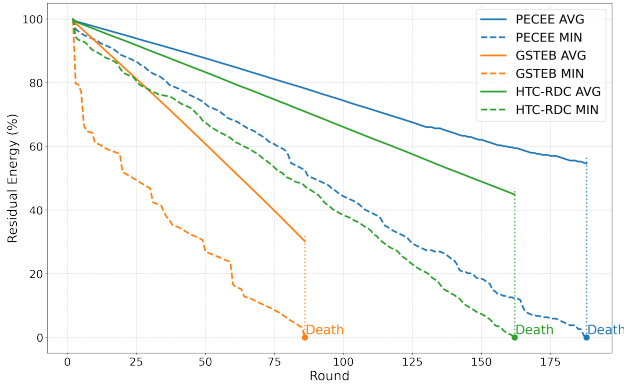


Fig. 7. Node residual energy over time for raw data scenario in a network of 400 nodes divided into 12 clusters.

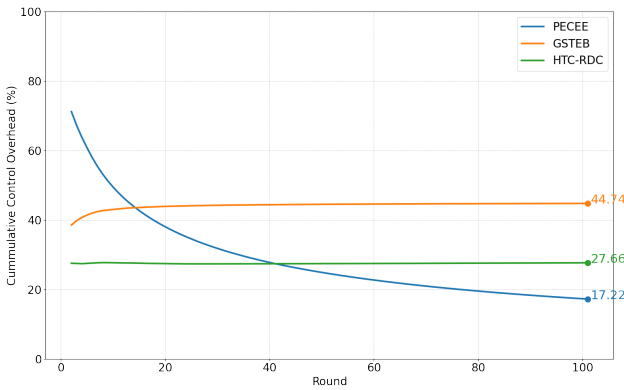


Fig. 8. Cumulative Control Overhead at round i .

showed that **for 400-node networks with large individual clusters, PECEE nearly triples the lifetime of GSTEB and doubles that of HTC-RDC²**. Initially, PECEE's high initialization cost(71%) reduces early performance, but control overhead drops(to 5.3%) by round 100, outperforming HTC-RDC(27%) and GSTEB(45%). This highlights a key insight: PECEE incurs substantial cost during initialization, however, once CGs and CFTs are fully established and reused intensively, the per-round overhead becomes negligible(as observed in Fig 8), enabling PECEE to extend network lifetime effectively.³ Table III shows that **with data aggregation, PECEE outperforms GSTEB, maintaining 200% network lifetime improvement in 400-nodes network**, and 30% in the smaller one.

VI. CONCLUSION

PECEE's Cellular Overlay reuses precomputed local topologies after each CH rotation, reducing control energy and avoiding full network reconfiguration. Though the initial Network Setup Phase incurs notable overhead, this cost

yields long-term gains and realizes PECEE's philosophy of *topology reuse*: trading early setup cost for sustained efficiency and stability. PECEE's Cellular Overlay reuses local topologies after CH rotation, reducing control energy and avoiding full network reconfiguration. Despite high initial setup costs, topology reuse ensures long-term efficiency. The framework scales effectively across network sizes, excelling in raw data applications like Precision Agriculture. Replacing simple heuristics with edge-AI could enable adaptive cluster optimization, supporting dynamic QoS demands while maintaining PECEE's scalability and energy efficiency for next-generation WSNs.

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²Line 2, Table II.

³In one example simulation: 43,470 packets in total for Initial Network Setup phase, and drops to 700(2–7 per node) packets per round

B. CONFERENCE ACCEPTANCE NOTIFICATION

This appendix provides the official acceptance notification confirming the acceptance of the paper included in Appendix A.

1/17/26, 10:03 PM

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Outlook

[WCNC 2026 Track 2] Your paper #1571218523 ('An Elastic Clustering Framework for Large-Scale WSNs Maximizing Network Lifetime') has been ACCEPTED!

From EDAS Conference Manager <help@edas.info>

on behalf of

WCNC 2026 Track 2 <wcnc2026track2-chairs@edas.info>

Date Wed 1/14/2026 9:40 PM**To** Khanh-Van Nguyen <vannk@soict.hust.edu.vn>; Quan A Le <anhquan.le52@gmail.com>; Vu Duc An 20215174 <An.VD215174@sis.hust.edu.vn>

Dear Dr. Khanh-Van Nguyen,

Congratulations! We are pleased to inform you that your paper #1571218523 "An Elastic Clustering Framework for Large-Scale WSNs Maximizing Network Lifetime" has been ACCEPTED for presentation at WCNC 2026 Track 2. Less than 50% have been accepted out of the more than 1000 papers submitted to IEEE WCNC 2026 Tracks.

Review reports of your paper are shown at the end of this email and are also available on EDAS at <https://edas.info/showPaper.php?m=1571218523>.

Information about the format of your presentation, date and time of your presentation, and the full program schedule will be announced later.

Accepted and presented papers will be published in the IEEE WCNC 2026 Conference Proceedings and submitted to IEEE Xplore® (subject to the fulfilment of all conditions provided below, as specified by IEEE ComSoc Policy).

==> PLEASE READ ALL INFORMATION GIVEN BELOW THOROUGHLY.

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Per IEEE ComSoc Policy, for each accepted paper to be published in the IEEE WCNC 2026 Proceedings and IEEE Xplore®, at least one author (including students) is required to register for the conference at either the FULL OR LIMITED AUTHOR REGISTRATION RATE. Moreover, the paper must be presented by an author named on the paper. Substitute presenters due to unforeseen circumstances must be authorized in advance by the TPC Chairs.

Non-refundable registration fees must be paid prior to uploading the final camera-ready version of the paper. Author registration will be opened on the IEEE WCNC 2026 website soon (<https://wcnc2026.ieee-wcnc.org/registration>). The deadline for registration is 7 February 2026

For authors with multiple accepted papers, one Full or Limited registration is valid for up to 3 (THREE) papers. Although the registration confirmation code enables submission of up to three papers, it can be used only once by the registered author, to create an author account for his/her submissions. Please do

APPENDIX B. CONFERENCE ACCEPTANCE NOTIFICATION

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not share your registration confirmation code with anyone else.

PAPER REVISION AND FINAL MANUSCRIPT UPLOAD

The final version of accepted papers (ready for publication, camera-ready) must be formatted as the standard 2-column IEEE layout and limited to 6 pages (or no more than 8 pages, by paying USD 100 over-length fee for each extra page).

You are strongly recommended to revise your paper in consideration of the reviewers' comments. Your Track Co-Chairs may contact you if they are not satisfied with your final version of your paper. Please note that the paper title as well as the list and order of authors CANNOT be changed at this stage. Failure to abide by this policy may result in the removal of your paper from the final conference program.

The upload of the final camera-ready manuscript is due by 7 February 2026. You must be registered for IEEE WCNC 2026 to upload your paper. Once you have registered for the conference, you will receive a confirmation code that you will use when you upload the final paper (by way of the upload link on EDAS).

Further details on the camera-ready manuscript preparation and upload process will be posted will be posted at the IEEE WCNC 2026 website (<https://wcnc2026.ieee-wcnc.org/submission-guidelines>).

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IEEE WCNC 2026 will feature sessions based on either oral lecture-style presentations (classroom) or interactive-style presentations (open space). This distinction is not related to the quality of the accepted papers. Papers will be assigned to sessions solely on the base of their topic. You will receive a separate email, informing you about the oral or interactive format of your presentation and the session, to which your paper will have been assigned. Please note that by submitting a manuscript all authors have implicitly agreed to present their accepted papers, regardless of the format, in sessions organized by the Conference Program Committee. Authors may not request to change the presentation format of their accepted papers.

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To be published in the IEEE WCNC 2026 Conference Proceedings and to be eligible for publication in IEEE Xplore®, an author of an accepted paper is required to register for the conference at the AUTHOR (member or non-member) rate and the paper must be presented by an author of that paper at the conference unless the TPC Chair grants permission for a substitute presenter arranged in advance of the

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event and who is qualified both to present and answer questions. Non-refundable registration fees must be paid prior to uploading the final IEEE formatted, publication-ready version of the paper. For authors with multiple accepted papers, one full registration is valid for up to 3 papers. Accepted and presented papers will be published in the IEEE WCNC 2026 Conference Proceedings and submitted to IEEE Xplore® as well as other Abstracting and Indexing (A&I) databases.

VISAS AND TRAVELLING TO KL

We recommend to make your travel arrangements and book your flight to KL well in advance. If you need a Visa Letter of Invitation from IEEE Communications Society (ComSoc), please follow instructions posted at: <https://wcnc2026.ieee-wcnc.org/visa-information>

Congratulations again for having your paper accepted in IEEE WCNC 2026, a flagship conference of the IEEE Communications Society. We look forward to welcoming you to Kuala Lumpur, Malaysia, in April 2026.

Thank you and best regards.

IEEE WCNC 2026 Track 2 Co-Chairs
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REVIEWERS' COMMENTS

===== WCNC 2026-Review 1 =====

> *** Relevance and Timeliness: Rate the importance and timeliness of the topic addressed in the paper within its area of research.

Acceptable. (3)

> *** Technical Content and Scientific Rigour: Rate the technical content of the paper (e.g.: completeness of the analysis or simulation study, thoroughness of the treatise, accuracy of the models, etc.), its soundness and scientific rigour.

Valid work but limited contribution. (3)

> *** Novelty and Originality: Rate the novelty and originality of the ideas or results presented in the paper.

Some interesting ideas and results on a subject well investigated. (3)

> *** Quality of Presentation: Rate the paper organization, the clearness of text and figures, the completeness and accuracy of references.

Readable, but revision is needed in some parts. (3)

> *** Strong Aspects: Comments to the author: What are the strong aspects of the paper?

The work addresses a significant and timely challenge: balancing the benefits of centralized cluster head (CH) selection with the overhead of frequent topology reconfiguration. The proposed solution is well-motivated.

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> *** Weak Aspects: Comments to the author: What are the weak aspects of the paper?

1. Irregular writing style: Many abbreviations are not spelled out upon first use, while some abbreviations (such as WSN) are repeatedly spelled out in full. For example, when GSTEB is first used, its full spelling and citation are not provided.
2. Some figures are unclear, such as Figure 1.
3. The paper lacks a relatively comprehensive description of the network model, such as node energy consumption, transmission radius, etc.
4. The paper mainly focuses on solving issues such as cluster construction and rotation, as well as data transmission in data transfer, but a large portion of the discussion is devoted to cell-related problems.

> *** Recommended Changes: Recommended changes. Please indicate any changes that should be made to the paper if accepted.

The above mentioned questions should be addressed.

> *** Submission Policy: Does the paper list the same author(s), title and abstract (minor wording differences in the abstract are okay) in its PDF file and EDAS registration? (yes/no)

yes

===== WCNC 2026-Review 2 =====

> *** Relevance and Timeliness: Rate the importance and timeliness of the topic addressed in the paper within its area of research.

Good. (4)

> *** Technical Content and Scientific Rigour: Rate the technical content of the paper (e.g.: completeness of the analysis or simulation study, thoroughness of the treatise, accuracy of the models, etc.), its soundness and scientific rigour.

Solid work of notable importance. (4)

> *** Novelty and Originality: Rate the novelty and originality of the ideas or results presented in the paper.

Significant original work and novel results. (4)

> *** Quality of Presentation: Rate the paper organization, the clearness of text and figures, the completeness and accuracy of references.

Well written. (4)

> *** Strong Aspects: Comments to the author: What are the strong aspects of the paper?

The paper presents an approach to maximize network lifetime in large-scale wireless sensor networks. It is clearly written, and the proposed solution is well described and supported by a pseudocode implementation.

> *** Weak Aspects: Comments to the author: What are the weak aspects of the paper?

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The proposed solution assumes that nodes have knowledge of their own position. While this may constitute a significant limitation, it is taken for granted in the paper, and no further details are provided regarding the positioning requirements.

> *** Recommended Changes: Recommended changes. Please indicate any changes that should be made to the paper if accepted.

Some aspects could be further improved. For instance:

- In Section C, it is stated that "Using GPS, nodes determine their coordinates and identify their corresponding cell within the hexagonal grid." Is the impact of GPS usage on the energy consumption of network nodes taken into account? If so, it would be useful to report this impact separately.

- Also in Section C, the authors state that "A node closest to the cell's center is designated as the Cell Leader (CL)." Does the distance between the selected node and the actual cell center affect the algorithm's performance? If so, a discussion or quantitative analysis would be valuable.

- The authors claim that the proposed solution is particularly suitable for scenarios such as Precision Agriculture, Intelligent Traffic Management, and Smart Hospital IoT Networks. However, GPS may perform poorly in dense urban environments and may not work at all indoors (e.g., in hospitals). Therefore, the authors should clarify the dependence of their solution on GPS. Is GPS mandatory, or can alternative positioning technologies be used? If not mandatory, what level of positioning accuracy is required for the proposal to function effectively? Does the location service to be always active?

> *** Submission Policy: Does the paper list the same author(s), title and abstract (minor wording differences in the abstract are okay) in its PDF file and EDAS registration? (yes/no)

Yes

===== WCNC 2026-Review 3 =====

> *** Relevance and Timeliness: Rate the importance and timeliness of the topic addressed in the paper within its area of research.

Good. (4)

> *** Technical Content and Scientific Rigour: Rate the technical content of the paper (e.g.: completeness of the analysis or simulation study, thoroughness of the treatise, accuracy of the models, etc.), its soundness and scientific rigour.

Solid work of notable importance. (4)

> *** Novelty and Originality: Rate the novelty and originality of the ideas or results presented in the paper.

Some interesting ideas and results on a subject well investigated. (3)

> *** Quality of Presentation: Rate the paper organization, the clearness of text and figures, the completeness and accuracy of references.

Readable, but revision is needed in some parts. (3)

> *** Strong Aspects: Comments to the author: What are the strong aspects of the paper?

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This paper proposes an elastic clustering framework named PECEE for large-scale wireless sensor networks, aiming to reduce cluster head rotation overhead and extend network lifetime through a "topology reuse" mechanism. Overall, the paper presents several interesting ideas and is generally well-structured.

> *** Weak Aspects: Comments to the author: What are the weak aspects of the paper?

- 1) The system model is based on some strong assumptions.
- 2) Simulations could be extended to more realistic scenarios.

> *** Recommended Changes: Recommended changes. Please indicate any changes that should be made to the paper if accepted.

- 1) It is better to provide more details on how the cells are formed, how to color the cells, how to autonomously select association relationships for grouping, and how "pending links" are established.
- 2) The core mechanisms, including "cell grid construction," "pending link configuration," and "cell path generation," are roughly described with the figures 2–4 only. It is not clear how these mechanisms are implemented.
- 3) The system model proposed in Section III-A is based on strong assumptions such as node homogeneity, and regular sparse deployment. While this offers some insights, it also diverges from common challenges encountered in real-world WSN deployments—such as unstable links, heterogeneous node capabilities, and irregular topologies. This makes the simulation results possibly unable to fully reflect the performance of PECEE in real environments.
- 4) The control overhead is discussed and analyzed in Figure 8. Why does PECEE incur significant costs during the initialization phase? Does it impact the network overhead?

> *** Submission Policy: Does the paper list the same author(s), title and abstract (minor wording differences in the abstract are okay) in its PDF file and EDAS registration? (yes/no)

Yes

===== WCNC 2026-Review 4 =====

> *** Relevance and Timeliness: Rate the importance and timeliness of the topic addressed in the paper within its area of research.

Good. (4)

> *** Technical Content and Scientific Rigour: Rate the technical content of the paper (e.g.: completeness of the analysis or simulation study, thoroughness of the treatise, accuracy of the models, etc.), its soundness and scientific rigour.

Solid work of notable importance. (4)

> *** Novelty and Originality: Rate the novelty and originality of the ideas or results presented in the paper.

Significant original work and novel results. (4)

> *** Quality of Presentation: Rate the paper organization, the clearness of text and figures, the completeness and accuracy of references.

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Well written. (4)

> *** Strong Aspects: Comments to the author: What are the strong aspects of the paper?

The paper introduces PECEE, which is designed to enable elastic clustering in large-scale WSNs for applications requiring raw, non-aggregated data collection, such as precision agriculture, intelligent traffic monitoring, and smart hospitals. PECEE combines centralized cluster head selection and localized cell-level routing and topology reuse in order to reduce control overhead. Simulation results demonstrate substantial improvements in network lifetime compared with baseline schemes such as GSTEB and HTC-RD.

The authors make use of the concept of pending links that are pre-established inter-cell connectors for facilitating quick reconfiguration. This sounds like a good idea for rotating cluster heads without reconstructing cluster-level routing from scratch.

The authors also present strong simulation results that demonstrate significant improvements over GSTEB in some 400-node scenarios as well as lower cumulative control overhead after initial setup and improved energy stability over time.

> *** Weak Aspects: Comments to the author: What are the weak aspects of the paper?

The authors do not discuss the communication complexity and storage overhead of the proposed PECEE framework, which weakens the claim of improving scalability. A more detailed complexity and fault tolerance analysis will improve the quality of the paper

PECEE assumes that nodes know their exact coordinates. Furthermore, the unit disk model with uniform transmission range is utilized. These strong assumptions may not be true in real-life situations. A better justification of these assumptions is required.

> *** Recommended Changes: Recommended changes. Please indicate any changes that should be made to the paper if accepted.

Two points listed above about the complexity analysis and justification of the assumptions should be addressed by the authors.

> *** Submission Policy: Does the paper list the same author(s), title and abstract (minor wording differences in the abstract are okay) in its PDF file and EDAS registration? (yes/no)

Yes.